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A Regime-Aware Hybrid GARCH-LSTM Ensemble Framework for Forecasting Nasdaq-100 Realized Volatility

Master's thesis

in the field of: Quantitative Finance

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Statements of the Supervisor

I declare that this thesis was prepared under my supervision and state that it fulfills the requirements for submission in the procedure for awarding a professional degree.

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During the preparation of this thesis, the author used generative artificial intelligence (AI) tools for language editing, proofreading LaTeX formatting, and code refinement.

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Summary

This thesis investigates hybrid and regime-aware approaches for forecasting Nasdaq-100 realized volatility. The study evaluates several GARCH-family models to select the best-performing specifications. Two main frameworks are developed: a hybrid LSTM model incorporating forecasts from the top three GARCH models as features, and a regime-aware ensemble combining the single best GARCH model and a standalone LSTM using regime probabilities estimated by a Gaussian Mixture Model. Model performance is evaluated using out-of-sample metrics and statistical tests against baseline benchmarks. The findings provide empirical evidence on integrating econometric models, deep learning, and market regime information for volatility forecasting.

Keywords

Volatility Forecasting; GARCH; LSTM; Hybrid Models; GMM; Regime Switching; Nasdaq-100

Field of the thesis (codes according to the Erasmus program)

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Hybrydowy model zespołowy GARCH–LSTM uwzględniający reżimy rynkowe w prognozowaniu zmienności realizowanej indeksu Nasdaq-100

Contents

1 INTRODUCTION

Financial market volatility plays a fundamental role in portfolio management, derivative pricing, risk management, and investment decision-making. Accurate volatility forecasts enable financial institutions and investors to assess market risk more effectively, optimize asset allocation, and improve hedging strategies. However, volatility exhibits complex characteristics, including time-varying persistence, clustering, nonlinear dynamics, and structural changes during periods of market stress. Although traditional econometric models, such as the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) family, have been widely applied to volatility forecasting, they may struggle to capture complex nonlinear relationships. Conversely, deep learning models, such as Long Short-Term Memory (LSTM) networks, have demonstrated strong predictive capabilities but often lack the interpretability and theoretical foundations of econometric approaches. Consequently, integrating both methodologies has become an increasingly important area of research.

The primary objective of this study is to evaluate whether combining traditional econometric models with deep learning techniques can improve the forecasting of Nasdaq-100 realized volatility. To achieve this objective, two approaches are investigated. The first develops a hybrid LSTM model that incorporates volatility forecasts generated by the three best-performing GARCH-family models as additional input features. The second constructs a regime-aware ensemble model that combines the forecasts of the best-performing GARCH model and the standalone LSTM model, with market conditions identified using a Gaussian Mixture Model (GMM). The GMM estimates the probabilities of calm and crisis regimes, which are subsequently incorporated into the ensemble forecasting process. In addition to these proposed approaches, three conventional forecasting methods—Historical Average, Simple Moving Average, and Simple Exponential Smoothing—are considered as baseline benchmarks, alongside the standalone GARCH-family and LSTM models. Using daily Nasdaq-100 market data and realized volatility estimated with the Yang-Zhang volatility estimator, all models are evaluated through out-of-sample forecasting. Forecasting performance is assessed using MAE, RMSE, and QLIKE, while the Diebold-Mariano test is employed to determine whether differences in predictive accuracy between competing models are statistically significant.

The study examines the following research hypotheses:

H1: The developed volatility forecasting models provide improved out-of-sample forecasting performance relative to the benchmark models.

H2: The standalone LSTM model provides superior out-of-sample volatility forecasts compared with the GARCH-family models.

H3: Incorporating forecasts from multiple GARCH-family models as additional inputs improves the out-of-sample forecasting performance of the standalone LSTM model.

H4: The regime-aware ensemble that combines the best-performing GARCH model and the standalone LSTM using GMM-derived regime probabilities provides superior out-of-sample volatility forecasting performance compared with the hybrid LSTM model.

The hypotheses stated above guide the empirical analysis presented in the subsequent chapters. Chapter 2 presents the literature review on financial volatility forecasting, covering GARCH-family models, deep learning approaches, hybrid forecasting methods, regime-based techniques, and conventional baseline forecasting methods. It discusses the theoretical and empirical foundations of the methodologies used in this study and identifies the research gap addressed by the study.

Chapter 3 describes the data used in the empirical analysis, including the data sources, variable definitions, data transformations, and descriptive statistics. It also presents the construction of the realized volatility measure and examines the statistical properties and stationarity of the variables used in the forecasting models. Finally, relevant preliminary diagnostic tests are conducted to assess the characteristics of the data and provide a basis for the subsequent model development. Chapter 4 presents the methodology and model specifications, including the conventional baseline models, GARCH-family models, LSTM networks, Gaussian Mixture Models, and ensemble forecasting approaches. It describes the data preparation, model estimation, forecasting procedure, and evaluation methods used to assess the forecasting performance of the competing models.

Chapter 5 presents and discusses the empirical results of the study, including the evaluation and comparison of the forecasting performance of the competing models. It reports the out-of-sample forecasting results using the selected evaluation metrics and examines whether the proposed hybrid and regime-aware approaches provide improvements over the conventional benchmarks and standalone models. Statistical comparisons are also conducted to assess whether the observed differences in forecasting accuracy are statistically significant. Finally, Chapter 6 concludes the study by revisiting the hypotheses formulated in the Introduction and discussing whether the empirical findings support or contradict them. It also relates the findings to the existing literature, assessing whether they are consistent with or differ from previous research.

2 LITERATURE REVIEW

2.1 State of Research and Methodological Foundations in Financial Volatility Forecasting

Volatility forecasting remains a cornerstone of quantitative finance, risk management, and asset allocation. Traditional econometric frameworks, particularly the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) family, have long dominated volatility modeling due to their ability to capture stylized facts of financial time series such as volatility clustering and leptokurtosis (Bollerslev, 1986; Engle, 1982). To better capture the complex realities of financial markets, the classical GARCH framework has undergone substantial extensions, notably through asymmetric specifications such as the Exponential GARCH (EGARCH) and GJR-GARCH models, which successfully account for the leverage effect—where negative shocks disproportionately increase volatility compared to positive shocks of an equal magnitude (Nelson, 1991; Glosten, Jagannathan, and Runkle, 1993). Furthermore, because standard financial returns frequently exhibit heavy tails and excess kurtosis that Gaussian assumptions fail to capture, modern implementations frequently incorporate alternative innovation distributions, such as Student-t, Generalized Error Distribution (GED), and skewed variants, to significantly improve the accuracy of conditional variance estimates (Bollerslev, 1987). However, despite these sophisticated extensions, standard econometric models often struggle to capture non-linear dependencies, long-memory properties, and rapid structural changes in high-frequency and turbulent market data.

In response to these limitations, machine learning (ML) and deep learning methods, particularly Long Short-Term Memory (LSTM) networks, have gained considerable attention due to their ability to capture complex and nonlinear temporal dependencies without imposing strict a priori structural assumptions (Fischer & Krauss, 2018). More recently, the literature has also focused on hybrid and ensemble approaches that combine the strengths of econometric and machine learning models. Such approaches integrate the statistical rigor of econometric volatility models with the predictive flexibility of neural networks, for example by incorporating GARCH forecasts or estimated parameters into LSTM architectures, with the aim of improving forecasting accuracy (Kim & Won, 2018; Hu et al., 2020).

An important finding in the literature is that the relative forecasting performance of econometric and machine learning models may vary across market conditions. Traditional econometric volatility models, particularly the GARCH family, tend to perform competitively during relatively stable periods, when volatility persistence and conditional variance dynamics can be effectively captured by established econometric structures. In a study on agricultural commodity volatility in India, Manogna and Dharmaji (2025) demonstrate that although traditional GARCH-family models perform well under stable market conditions, their performance may deteriorate in the presence of abrupt external shocks. In contrast, machine learning approaches, including recurrent neural networks (RNNs), long short-term memory (LSTM) networks (Hochreiter and Schmidhuber, 1997), and gated recurrent units (GRUs) (J.Chung et al., 2014/2015), can capture complex nonlinear temporal dependencies and potentially adapt more flexibly to changing market conditions. This indicates that information about the prevailing volatility regime may provide useful information for combining forecasts from GARCH-family and machine learning models.

Incorporating estimated regime probabilities into the forecasting process therefore provides a framework for combining model forecasts while accounting for changes in market conditions. Rather than selecting one model exclusively, a probability-weighted ensemble can dynamically adjust the contribution of each model according to the estimated probability of the prevailing market regime. Under this approach, the GARCH-family forecast can receive greater weight when calm-market conditions are more likely, while the LSTM forecast can receive greater weight when crisis conditions are more probable.

Building on the motivation for a regime-aware forecasting framework, it is first necessary to identify market conditions under which forecasting models may exhibit different performance. Financial markets can alternate between distinct states, often characterized by relatively calm and high-volatility or crisis periods, suggesting that volatility dynamics may not be homogeneous over time. This motivates the use of unsupervised clustering methods such as Gaussian Mixture Models (GMM), which provide a flexible, data-driven approach to identifying latent market regimes based on the statistical characteristics of observed market variables. The suitability of this approach is further supported by the literature, which documents heterogeneous and regime-dependent dynamics in financial time series and demonstrates the usefulness of Gaussian mixture structures for representing such variation. Kalliovirta, Meitz, and Saikkonen (2016) develop a Gaussian Mixture Vector Autoregressive (GMVAR) model specifically for time series exhibiting regime-switching behaviour. Their framework allows different regimes to be associated with economically meaningful characteristics of the underlying process, demonstrating the suitability of Gaussian mixture structures for representing nonlinear and regime-dependent dynamics. Similarly, Bezerra and Albuquerque (2017) investigate volatility forecasting using SVR–GARCH models with mixtures of Gaussian kernels and find that models incorporating multiple Gaussian components improve out-of-sample volatility forecasts and are able to capture regime-switching behaviour in stock-market volatility. Further evidence for the relevance of regime-dependent volatility is provided by Zhang et al. (2019), who compare single-regime GARCH models with regime-switching GARCH specifications for WTI and Brent crude oil. Although their regime-switching models provide advantages in some in-sample settings, they do not consistently outperform single-regime models out of sample, highlighting that the usefulness of regime information may depend on how it is incorporated into the forecasting framework. Taken together, these studies motivate the use of GMM in the present study as a flexible, data-driven approach for identifying observations associated with different volatility states. Rather than imposing a specific parametric regime-switching structure on the volatility model, GMM is used to estimate the probability that the market belongs to each latent regime. These regime probabilities are subsequently incorporated into the ensemble forecasting framework alongside the GARCH and LSTM forecasts, allowing the contribution of each forecast to reflect the estimated likelihood of the prevailing market regime.

Recent research also highlights the importance of evaluating complex forecasting models against simple benchmark methods. In their evaluation of time series with low predictability, Beck et al. (2025) emphasize that no forecasting method consistently outperforms the naive (no-change) forecast for highly volatile time series from exchange-rate and stock-price datasets. Their findings suggest that, in settings with limited predictability, comparisons among increasingly complex models may be less informative when simple benchmarks are not considered. Similarly, Baráth et al. (2026) shows that both classical and advanced forecasting methods can achieve high forecasting accuracy, with the characteristics of the underlying data potentially playing a more important role than model complexity. Together, these findings reinforce the importance of including simple, adaptive benchmarks when assessing more sophisticated volatility forecasting models, as greater model complexity does not necessarily translate into superior out-of-sample performance. In this study, Historical Average, Simple Moving Average, and Simple Exponential Smoothing are therefore included as baseline models to provide reference points for assessing whether the additional complexity of GARCH-family, LSTM, hybrid, and regime-aware approaches leads to meaningful improvements in out-of-sample volatility forecasting accuracy.

Furthermore, the choice of volatility measure is an important consideration in volatility forecasting, as the latent volatility of financial assets cannot be directly observed. Consequently, empirical studies rely on observable volatility proxies to evaluate forecasting performance. Range-based estimators are particularly useful because they can exploit information contained in daily price ranges and provide a richer representation of volatility dynamics than measures based solely on closing prices. Yang and Zhang (2000) propose one such estimator, which has subsequently been applied in empirical volatility forecasting studies.

The methodology employed in this study draws on several established research streams in financial volatility forecasting. GARCH-family models provide the econometric foundation for modeling conditional volatility, while LSTM networks offer a flexible framework for capturing nonlinear and temporal dependencies. Previous empirical research on GARCH–LSTM models supports the integration of econometric volatility information with deep learning approaches. The study also draws on the literature on market regime identification and forecast combination, which provides the methodological basis for incorporating regime information and combining forecasts from different model classes. These research streams provide the theoretical and empirical foundations for the methodological choices adopted in the subsequent analysis.

2.2 Research Gap and Contribution

While the application of GARCH-family models and LSTM networks to equity indices such as the NASDAQ-100 is well documented, existing studies often evaluate individual models or hybrid specifications without directly comparing alternative mechanisms for integrating econometric and deep learning forecasts. In particular, limited attention has been given to whether GARCH-based information is more effectively incorporated as additional features within a deep learning architecture or combined with LSTM forecasts at the prediction level. This distinction is important because these approaches exploit information from the component models in fundamentally different ways.

Previous research has demonstrated that regime-switching and mixture-based approaches can capture heterogeneous volatility dynamics across different market states. However, much of this literature focuses on estimating separate volatility processes for different regimes rather than using probabilistic regime information to determine the relative contribution of forecasts from fundamentally different model classes. Consequently, it remains unclear whether identifying calm and crisis conditions through probabilistic regime classification can translate into additional out-of-sample forecasting gains when GARCH and LSTM forecasts are combined. This motivates a direct comparison between regime-aware forecast combination and feature-level integration within a single deep learning architecture.

A further limitation concerns the choice of benchmarks. Although recent volatility forecasting studies frequently compare deep learning models with GARCH-family specifications, relatively simple forecasting methods such as Historical Average, Simple Moving Average, and Simple Exponential Smoothing are not always given equal consideration. Consequently, demonstrating improvements over GARCH-family models alone does not necessarily establish that a complex forecasting model offers meaningful gains over simpler alternatives. A comprehensive evaluation against strong simple benchmarks is therefore necessary to determine whether additional model complexity translates into genuine out-of-sample forecasting gains.

This study addresses these gaps through two complementary methodological approaches:

1. **Sequential Hybrid versus Regime-Aware Ensemble:** The study provides a direct empirical comparison between two distinct approaches to integrating econometric and deep learning information. The study provides a direct empirical comparison of two approaches to integrating econometric and deep learning information at different stages of the forecasting process. Specifically, it examines whether integration at the feature level or at the prediction level provides greater predictive value for Nasdaq-100 realized volatility forecasting. This comparison offers empirical evidence on whether the stage at which information from different model classes is integrated affects out-of-sample forecasting performance.

2. **Probabilistic Regime Identification and Forecast Combination:** The study employs a Gaussian Mixture Model to identify calm and crisis market conditions and generate

probabilistic regime information. Furthermore, the study directly evaluates whether these probabilities generate additional out-of-sample forecasting gains when used in the ensemble. This provides an empirical test of whether regime information is useful not only for describing changing market conditions but also for improving the combination of forecasts from different model classes.

3. **Evaluation against Strong Simple Benchmarks:** The study evaluates all proposed approaches alongside Historical Average, Simple Moving Average, and Simple Exponential Smoothing. Including these simple benchmarks provides a broader basis for assessing whether the proposed models offer meaningful improvements in out-of-sample forecasting performance. The inclusion of these benchmarks is particularly important given the high volatility and limited predictability that characterize financial time series.

   Taken together, the study contributes to the existing literature by providing an integrated empirical comparison of parametric, deep learning, hybrid, and regime-aware forecasting approaches, alongside simple baseline models, under a common out-of-sample evaluation framework. By systematically assessing the predictive contribution of model complexity and regime information against these benchmarks, the analysis examines whether additional modelling elements translate into measurable forecasting gains. The resulting evidence therefore provides a more nuanced assessment of when and how econometric forecasts and regime information contribute to volatility forecasting, including in combination with deep learning-based models.

3 DATA SOURCES AND COLLECTION METHOD

3.1 Data Sources and Study Period

The empirical analysis uses daily market data for the Nasdaq100 index from 1 January 2000 to 31 December 2025, comprising a total of 6,538 daily observations. The data, downloaded from Yahoo Finance, include daily open, high, low, and close prices, together with trading volume. The CBOE Nasdaq100 Volatility Index (VXN), also obtained from Yahoo Finance, is incorporated to capture information about market expectations of future volatility.

The 26-year sample provides a sufficiently large number of observations for training the LSTM model and developing a robust out-of-sample forecasting framework. More importantly, the long sample covers a wide range of market conditions, including both relatively calm periods and episodes of substantial market turbulence. This variation provides the necessary basis for identifying distinct market regimes using the Gaussian Mixture Model (GMM) and subsequently incorporating the estimated regime probabilities into the forecast-combination process. The observations are maintained in chronological order throughout the analysis to preserve the time-series structure and prevent information from the future from entering the forecasting process.

3.2 Study Population and Sample Selection

The study population covers all available trading days of the NASDAQ-100 index within the defined sample period. To ensure a consistent and unbiased comparison of forecasting performance, the dataset is divided sequentially into three non-overlapping subsets: a training set, a validation set, and a test set. The training set is used for model estimation and feature preparation, the validation set for hyperparameter tuning and model selection, and the test set is reserved exclusively for out-of-sample evaluation. This identical temporal partition is applied consistently across all models, ensuring that every forecasting approach is estimated, tuned, and evaluated using the same information sets and time periods. The chronological structure of the split also prevents look-ahead bias by ensuring that information from future periods is not used during model estimation or hyperparameter selection.

The resulting data partition is as follows:

- Training Set (80%): 4 January 2000 – 14 October 2020 (5229 observations)
- Validation Set (10%): 15 October 2020 – 22 May 2023 (654 observations)
- Test Set (10%): 23 May 2023 – 30 December 2025 (654 observations)

3.3 Variable Definitions

To support the multi-stage empirical framework, which incorporates econometric estimation, deep learning, regime identification, and forecast combination, the variables are organized into four functional categories according to their role within the modelling pipeline:

1. **Base Market Variables and Volatility Target:** This category comprises the raw market data, derived return measures, and Yang–Zhang realized volatility, which serves as the proxy for the unobserved true volatility and the primary forecasting target.

2. **Econometric Model Variables and Forecasts:** This category includes the volatility forecasts generated by the GARCH-family models evaluated in Stage 1. The three best-performing GARCH-family models are selected and their forecasts are subsequently incorporated as input features in the Hybrid LSTM model. Among these models, the single best-performing GARCH model is additionally selected for use in the final ensemble framework.

3. **Machine Learning Features:** The standalone LSTM model uses rolling historical volatility measures as input features. The Hybrid LSTM extends these inputs by incorporating the forecasts from the three selected GARCH-family models, thereby combining historical volatility information with forecasts generated by econometric models.

4. **Regime and Ensemble Variables:** This category includes the calm- and crisis-regime probabilities estimated using the Gaussian Mixture Model (GMM) in Stage 4. These probabilities are used in the final ensemble framework to dynamically combine the forecasts from the single best-performing GARCH model and the standalone LSTM model.

Table 1 presents the variables used in the study, together with their corresponding mathematical symbols and units of measurement.

The realized volatility measures used throughout the empirical analysis are constructed from the daily Yang–Zhang realized volatility proxy. The one-day realized volatility is given directly by the daily measure:

$$RV_{1d,t} = RV_{t-1} \tag{1}$$

To capture volatility over longer horizons, 5-day and 22-day realized volatility measures are constructed from the corresponding daily realized volatility observations. The 5-day measure is defined as:

$$RV_{5d,t} = \frac{1}{5}\sum_{i=0}^{4} RV_{t-1-i} \tag{2}$$

while the 22-day measure is defined as:

Table 1: Variable Definitions, Mathematical Notation, and Units of Measurement

Variable Name	**Symbol**	**Unit of Measurement**
Open Price	O_t	Index Points
High Price	H_t	Index Points
Low Price	L_t	Index Points
Close Price	C_t	Index Points
Log Returns	r_t	Percentage Points (%)
Yang–Zhang Realized Volatility	RV_t	Daily Percentage Points (%)
Log NASDAQ-100 Volatility Index	$\ln(\mathrm{VXN}_t)$	Log-Index Points
Log Parkinson Volatility	$\ln(\sigma_{\mathrm{Parkinson},t})$	Log-Percentage
1-Day Realized Volatility	$RV_{1\mathrm{d},t}$	Daily Percentage Points (%)
5-Day Realized Volatility	$RV_{5\mathrm{d},t}$	Daily Percentage Points (%)
22-Day Realized Volatility	$RV_{22\mathrm{d},t}$	Daily Percentage Points (%)
5-Day EMA Volatility	$\mathrm{EMA}_{5\mathrm{d},t}$	Daily Percentage Points (%)
22-Day EMA Volatility	$\mathrm{EMA}_{22\mathrm{d},t}$	Daily Percentage Points (%)
5-Day Rolling Return Standard Deviation	$\sigma_{5\mathrm{d},t}$	Percentage Points (%)
22-Day Rolling Return Standard Deviation	$\sigma_{22\mathrm{d},t}$	Percentage Points (%)

$$RV_{22d,t} = \frac{1}{22}\sum_{i=0}^{21} RV_{t-1-i} \tag{3}$$

3.4 Data Transformations and Cleaning Procedures

Prior to model estimation, the raw daily Nasdaq-100 index and VXN data are examined for missing observations. Non-trading days, including weekends and exchange holidays, are naturally excluded from the dataset. No extreme observations are removed from the sample, as periods of unusually high returns and volatility represent genuine market conditions and contain important information for volatility forecasting and regime identification.

Stationarity tests are conducted on the relevant variables before model estimation to assess their time-series properties. The non-stationary price series is transformed into daily logarithmic returns which are subsequently used for GARCH-family model estimation. The detailed results of the stationarity tests are presented in the descriptive and statistical analysis section.

For the machine learning models and GMM, the input features are standardized using scaling parameters estimated exclusively from the training data. The same transformation is subsequently applied to the validation and test observations to prevent information from future observations from entering the preprocessing stage and causing data leakage. To reduce scale-related effects and stabilize the variance of the inputs, the lagged volatility measures are transformed using the natural logarithm before being provided to the standalone LSTM and hybrid LSTM models. The target volatility variable is retained on its original scale for the final evaluation of the forecasts.

The dataset is divided into training, validation, and test sets, with each subset serving a specific role in model development and out-of-sample evaluation. The GARCH-family models and the GMM are estimated using the corresponding training, validation, and test periods, as defined above, while the LSTM and hybrid LSTM models use a rolling walk-forward procedure with a fixed-length window applied to the combined training and validation data. At each forecasting point, the models are re-estimated using only the most recent available observations within the fixed window, and the resulting forecasts are evaluated on the test set. This procedure preserves the temporal structure of the data and prevents information from future observations from entering the model estimation process.

4 DESCRIPTIVE ANALYSIS AND DIAGNOSTIC TESTS

4.1 Price, Return, and Volatility Dynamics

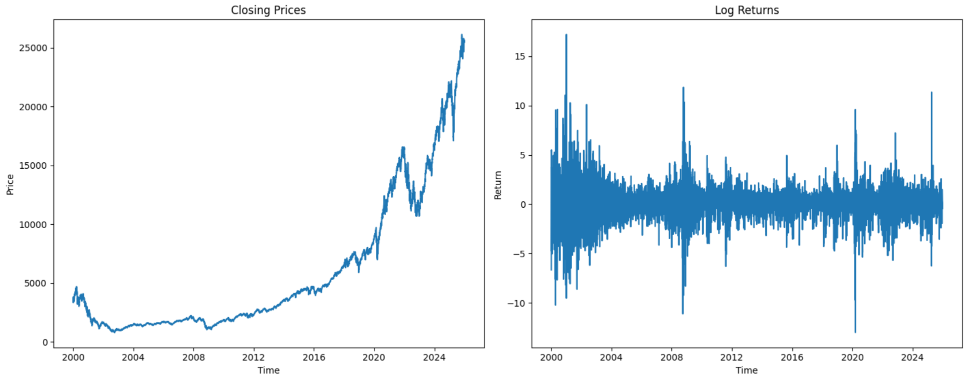

Figure 1: Closing Prices and Log Returns.

As shown in Figure 1, the closing price of NDX exhibits a clear non-stationary pattern over the sample period. In contrast, the log returns appear approximately stationary and exhibit volatility clustering, whereby periods of high volatility tend to be followed by further periods of high volatility. Moreover, the return series shows substantial fluctuations, including several pronounced positive and negative spikes, indicating the presence of extreme movements in the market.

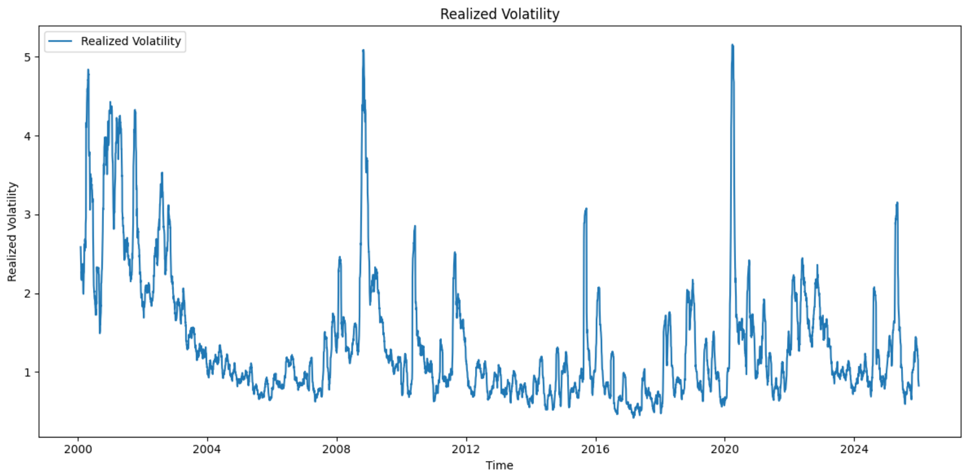

Figure 2: Realized Volatility Dynamics.

The realized volatility dynamics shown in Figure 2 further support the patterns observed in the price and return series. The NDX exhibits substantial variation in volatility over the sample period, with realized volatility reaching a maximum of approximately 5%.

4.2 Distributional Properties

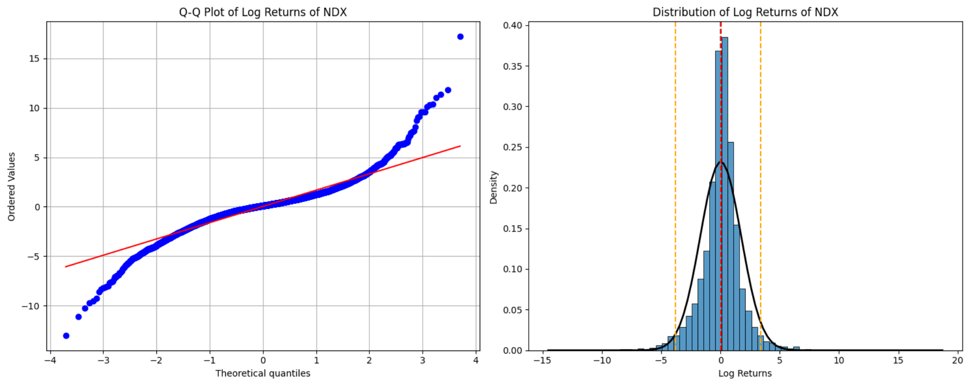

Figure 3: Distribution of NDX log returns.

As shown in Figure 3, both the density plot and Q–Q plot indicate that the distribution of log returns deviates substantially from normality. Compared with the normal distribution, the return series exhibits heavier tails on both the left and right sides, with observations extending further into the extremes. The two orange dashed lines in the right-hand panel, representing the 2.5% and 97.5% quantiles, highlight a significant number of observations lying outside these bounds. This indicates a higher frequency of large positive and negative returns than would be expected under a normal distribution and suggests the presence of fat-tailed behavior.

In addition, the Jarque–Bera test was conducted to formally assess the normality of the log-return distribution. The resulting p-value is very close to zero, providing strong evidence against the null hypothesis of normality and confirming the conclusion suggested by the density and Q–Q plots.

The descriptive statistics presented in Table 2 provide an initial overview of the statistical properties of the variables used in the analysis. The reported measures summarize their central tendency, dispersion, and distributional characteristics, providing a preliminary assessment of the behavior of the data prior to model estimation.

Table 2: Summary Statistics: Realized Volatility vs. Log Returns

Statistic	**Volatility**	**Log Returns**
Mean	1.37	0.03
Std Dev	0.81	1.72
Min	0.42	-13.00
Max	5.16	17.20
Skewness	1.94	0.05
Kurtosis	4.02	7.17

The skewness of the log returns is 0.05, indicating that the return distribution is approximately symmetric around its mean, with only a slight positive skew. In contrast, the kurtosis of the log returns is relatively high at 7.17, indicating substantial tail heaviness and a greater occurrence of extreme observations than would be expected under a normal distribution. This finding is consistent with the graphical and Jarque–Bera normality assessments presented above. Furthermore, the standard deviation of the log returns is considerably larger than their mean, highlighting the substantial variability of daily returns relative to their average level. The realized volatility series also exhibits high kurtosis, providing further evidence of fat-tailed behavior and the presence of periods with unusually high volatility.

In general, the initial examination of the data supports the use of alternative innovation distributions with heavier tails in the subsequent volatility models.

4.3 Stationarity tests

The stationarity of the variables is examined using the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests, with the results reported in Table 3. Both tests have a null hypothesis of a unit root, such that rejection of the null indicates that the series is stationary. The ADF test accounts for serial correlation by including lagged differences of the dependent variable, while the PP test applies a non-parametric correction for serial correlation and heteroskedasticity. Using both tests provides a more robust assessment of the stationarity properties of the variables.

Table 3: Unit Root Tests: Log Returns vs. Realized Volatility

Var	**ADF Stat**	**ADF Critical Value**	**PP Stat**	**PP Critical Value**
Log Returns	-15.389	-2.567 (1%)	-87.984	-3.43 (1%)
Realized Volatility	-2.118	-1.941 (5%)	-5.786	-3.43 (1%)

The number of lags selected for the log returns and realized volatility series is 25 and 89, respectively. The relatively large number of lags for realized volatility may be attributed to the strong serial correlation and persistence present in the series. At the corresponding significance levels, both the ADF and PP tests provide evidence that the log returns and realized volatility series are stationary. Therefore, both series can be used as inputs to subsequent forecasting models without concerns about spurious regression relationships arising from non-stationarity.

4.4 Autocorrelation and Conditional Heteroskedasticity

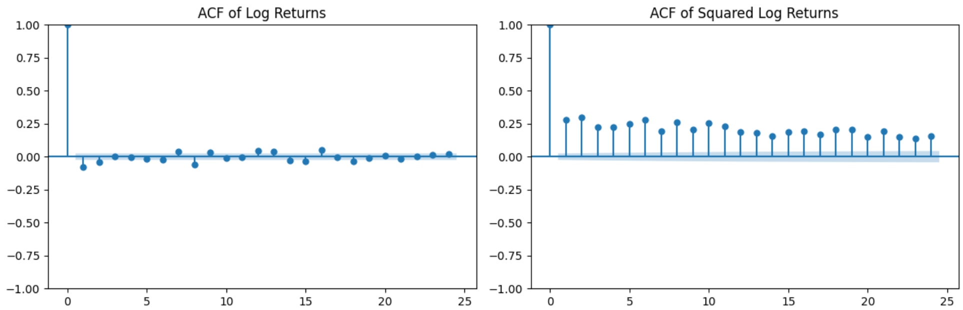

Figure 4: Autocorrelation Structure of Log Returns and Squared Log Returns.

As shown in Figure 4, the ACF of log returns exhibits one statistically significant spike at the first lag, suggesting that a simple AR(1) specification may be appropriate for modeling the conditional mean of the return series. In contrast, the ACF of squared log returns shows statistically significant autocorrelation across multiple lags, providing evidence of serial dependence in the variance and supporting the presence of volatility clustering.

The Engle ARCH-LM test is therefore employed to formally examine the presence of autoregressive conditional heteroskedasticity in the return series. The null hypothesis states that there are no ARCH effects, while rejection of the null provides evidence of conditional heteroskedasticity.

Table 4: ARCH LM Test for Conditional Heteroskedasticity

Statistic	**Value**
LM Statistic	1073.417
LM p-value	0.000
F-Statistic	256.664
F p-value	0.000

The results presented in Table 4 confirm the presence of significant ARCH effects, as indicated by both the LM and F-statistics. The corresponding p-values are very close to zero, leading to a strong rejection of the null hypothesis of no ARCH effects. This finding offers further empirical support for the use of GARCH-family models to capture the time-varying volatility dynamics of the return series.

5 DESCRIPTION OF THE ECONOMETRIC MODELS AND FORECASTING TECHNIQUES

5.1 Baseline Forecasting Models

To provide simple reference points for evaluating the more complex volatility forecasting models, three conventional forecasting methods are considered as baseline models: the Historical Average (HA), Simple Moving Average (SMA), and Simple Exponential Smoothing (SES). These models use only past observations of the realized volatility measure and provide relatively straightforward representations of the underlying volatility process. These models differ in the way past observations are weighted when generating the one-step-ahead volatility forecast.

5.1.1 Historical Average

The Historical Average model forecasts future volatility using the arithmetic mean of the realized volatility observations available at each forecasting point. Under the walk-forward forecasting procedure, the one-step-ahead forecast at time t is given by:

$$RV_{t+1|t} = \frac{1}{t}\sum_{i=1}^{t} RV_i \tag{4}$$

The estimation sample is updated sequentially as new observations become available, while equal weight is assigned to all observations included up to the forecasting point. Consequently, the forecast represents the historical average level of volatility and does not explicitly account for changes in volatility persistence or short-term dynamics. The Historical Average therefore provides a simple benchmark against which more flexible forecasting models can be evaluated.

5.1.2 Simple Moving Average

The Simple Moving Average (SMA) model assigns equal weight to a fixed number of the most recent realized volatility observations. Under the walk-forward forecasting procedure, the one-step-ahead forecast at time t, using a window of length m, is defined as:

$$RV_{t+1|t} = \frac{1}{m}\sum_{i=0}^{m-1} RV_{t-i} \tag{5}$$

In this study, the window length is set to m=22, corresponding approximately to one trading month. At each forecasting point, the estimation window moves forward by one observation, incorporating the most recently available realized volatility and excluding the oldest observation in the window. Compared with the Historical Average, the SMA places greater emphasis on recent market conditions by restricting the estimation sample to the selected window. The 22-day window provides a balance between responsiveness to recent volatility changes and stability of the forecast. The SMA therefore provides a simple benchmark that captures short- to medium-term volatility dynamics without requiring a parametric volatility model.

5.1.3 Simple Exponential Smoothing

The Simple Exponential Smoothing (SES) model assigns progressively declining weights to older realized volatility observations, allowing the forecast to respond to recent changes in volatility. The level component is updated recursively as:

$$\ell_t = \alpha RV_t + (1-\alpha)\ell_{t-1}, \qquad 0 < \alpha < 1 \tag{6}$$

where ℓ_t denotes the estimated level of the volatility series and α is the smoothing parameter. The one-step-ahead forecast is given by:

$$RV_{t+1|t} = \ell_t \tag{7}$$

The smoothing parameter determines how quickly the forecast responds to new observations. A larger α places more weight on the most recent observation and therefore makes the forecast more responsive, while a smaller α produces a smoother forecast by giving greater weight to past observations. In this study, the smoothing parameter α is estimated using the available training and validation data through automatic parameter optimization. The parameter is determined by numerically minimizing the in-sample sum of squared errors (SSE) over its admissible range $0 < \alpha < 1$. The resulting value of α is then kept fixed during the walk-forward forecasting period, while the volatility level is updated sequentially as new observations become available.

The three baseline models provide complementary reference points. The Historical Average represents a long-run volatility level, the Simple Moving Average provides a measure of the underlying market volatility trend over a fixed historical window, and Simple Exponential Smoothing allows recent observations to receive greater weights while retaining information from the past. Their forecasts are subsequently evaluated using the same out-of-sample forecasting framework and loss measures as the more advanced models.

5.2 Realized Volatility Proxy: The Yang-Zhang Formula

The Yang–Zhang realized volatility estimator proposed by Yang and Zhang (2000) is used as the forecasting target. The estimator combines three components: overnight volatility, open-to-close volatility, and a Rogers–Satchell measure of intraday volatility.

The normalized log-price differences are defined as follows:

- **Overnight Return**:

$$r_{\text{overnight},t} = \ln(O_t) - \ln(C_{t-1}) \tag{8}$$

- **Open-to-Close Return**:

$$r_{\text{close},t} = \ln(C_t) - \ln(O_t) \tag{9}$$

- **Normalized High Price**:

$$r_{\text{high},t} = \ln(H_t) - \ln(O_t) \tag{10}$$

- **Normalized Low Price**:

$$r_{\text{low},t} = \ln(L_t) - \ln(O_t) \tag{11}$$

Over a rolling estimation window of N trading days, the Yang-Zhang realized variance estimator $\hat{\sigma}^2_{YZ,t}$ is expressed as a weighted sum of the overnight variance ($\sigma^2_{\text{overnight}}$), open-to-close variance (σ^2_{close}), and Rogers-Satchell variance (σ^2_{RS}):

$$\hat{\sigma}^2_{\text{YZ},t} = \sigma^2_{\text{overnight}} + k\,\sigma^2_{\text{close}} + (1-k)\,\sigma^2_{\text{RS}} \tag{12}$$

where individual components and sample variances over the N-day rolling window are defined by:

$$\sigma^2_{\text{overnight}} = \frac{1}{N-1}\sum_{i=0}^{N-1}\left(r_{\text{overnight},t-i} - \bar{r}_{\text{overnight}}\right)^2, \quad \bar{r}_{\text{overnight}} = \frac{1}{N}\sum_{i=0}^{N-1} r_{\text{overnight},t-i} \tag{13}$$

$$\sigma^2_{\text{close}} = \frac{1}{N-1}\sum_{i=0}^{N-1}\left(r_{\text{close},t-i} - \bar{r}_{\text{close}}\right)^2, \quad \bar{r}_{\text{close}} = \frac{1}{N}\sum_{i=0}^{N-1} r_{\text{close},t-i} \tag{14}$$

$$\sigma^2_{\text{RS}} = \frac{1}{N}\sum_{i=0}^{N-1}\left[r_{\text{high},t-i}\left(r_{\text{high},t-i} - r_{\text{close},t-i}\right) + r_{\text{low},t-i}\left(r_{\text{low},t-i} - r_{\text{close},t-i}\right)\right] \tag{15}$$

In this study, the estimation window is set to N=22 trading days, corresponding approximately to one month of trading activity.

The weighting parameter $k \in (0,1)$ optimizes the tradeoff between minimizing variance and handling continuous drift:

$$k = \frac{0.34}{1.34 + \frac{N+1}{N-1}} \tag{16}$$

The daily realized volatility proxy $\hat{\sigma}_{YZ,t}$ is obtained by taking the square root of the estimated Yang–Zhang realized variance and expressing the resulting volatility in percentage terms:

$$\hat{\sigma}^{\text{daily}}_{YZ,t} = 100\sqrt{\hat{\sigma}^2_{YZ,t}} \tag{17}$$

The resulting realized volatility measure is obtained as:

$$RV_t = \hat{\sigma}^{\text{daily}}_{YZ,t} \tag{18}$$

The Yang–Zhang estimator is selected because it makes use of the complete daily OHLC information available in the dataset rather than relying exclusively on closing prices. In particular, it accounts for overnight price movements and intraday price variation and is designed to reduce the influence of opening-price jumps and market microstructure effects on the volatility estimate.

5.3 GARCH-Family Models

The first stage of the empirical analysis evaluates alternative GARCH-family specifications for forecasting Nasdaq-100 volatility. Three conditional variance models are considered: GARCH, EGARCH, and GJR-GARCH. To account for different distributional characteristics of financial returns, each volatility specification is estimated under three alternative innovation distributions: the normal distribution, Student's (t) distribution, and skewed Student's (t) distribution. In addition, two alternative conditional mean specifications are considered: a constant mean and an ARMA(1,0) specification, which is equivalent to an AR(1) mean process. Consequently, the empirical analysis evaluates 18 candidate GARCH-family specifications.

Raw asset prices typically exhibit non-stationarity and stochastic trends, they are transformed into a continuously compounded log return series, r_t, defined as:

$$r_t = \ln(C_t) - \ln(C_{t-1}) = \ln\left(\frac{C_t}{C_{t-1}}\right) \tag{19}$$

The return equation is specified as

$$r_t = \mu_t + \varepsilon_t \tag{20}$$

where μ_t represents the conditional mean and ε_t is the innovation.

Two alternative specifications for the conditional mean are considered. The first assumes a constant conditional mean

$$\mu_t = \mu \tag{21}$$

while the second incorporates autoregressive dynamics

$$\mu_t = c + \sum_{i=1}^{p} \phi_i r_{t-i} \tag{22}$$

where c is a constant, ϕ_i are autoregressive coefficients, and p denotes the selected autoregressive order.

The innovation is expressed as

$$\varepsilon_t = \sigma_t z_t, \tag{23}$$

where σ_t is the conditional volatility and z_t is a standardized innovation with zero mean and unit variance. The conditional distribution of z_t is varied across the candidate specifications to account for alternative distributional characteristics of financial returns.

5.3.1 Model Specifications

GARCH Specification

For the GARCH(1,1) specification, the conditional variance is modeled as

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \tag{24}$$

where $\omega > 0$, $\alpha \geq 0$, and $\beta \geq 0$. The parameters α and β represent the effects of recent shocks and past conditional variance, respectively. Volatility persistence requires that $\alpha + \beta < 1$, which ensures covariance stationarity and a mean-reverting volatility process.

EGARCH Specification

The EGARCH specification models the logarithm of conditional variance and allows for asymmetric responses to positive and negative innovations. For the EGARCH(1,1) specification, the conditional variance is therefore modeled as:

$$\ln(\sigma_t^2) = \omega + \beta\ln(\sigma_{t-1}^2) + \alpha\left(|z_{t-1}| - \mathbb{E}[|z_{t-1}|]\right) + \gamma z_{t-1} \tag{25}$$

where $z_{t-1} = \frac{\varepsilon_{t-1}}{\sigma_{t-1}}$ is the standardized innovation, and $\mathbb{E}|z_{t-1}|$ is the expected value of the absolute standardized innovation. The parameter γ captures the asymmetric response of volatility to innovations. In particular, a non-zero value of γ allows positive and negative shocks of the same magnitude to have different effects on future volatility. Unlike the standard GARCH model, the logarithmic formulation ensures a positive conditional variance without imposing non-negativity restrictions directly on the variance parameters. For the EGARCH(1,1) specification, covariance stationarity generally requires $|\beta| < 1$.

GJR-GARCH Specification

The GJR-GARCH specification extends the standard GARCH model by incorporating an indicator variable that allows negative innovations to have an additional effect on conditional volatility. For the GJR-GARCH(1,1) specification, the conditional variance is modeled as:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma I_{\{\varepsilon_{t-1}<0\}}\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 \tag{26}$$

The indicator $I_{\{\varepsilon_{t-1}<0\}}$ takes the value one when the previous innovation is negative and zero otherwise. For the GJR-GARCH model, the parameters satisfy the standard GARCH constraints, with the additional restriction $\alpha + \gamma \geq 0$ to ensure a non-negative conditional variance. Under symmetric innovations, covariance stationarity additionally requires $\alpha + \beta + \gamma/2 < 1$.

5.3.2 Conditional Innovation Distributions and Candidate Specifications

Each of the three conditional variance specifications is estimated under three alternative distributions for the standardized innovation z_t: the normal, Student's (t), and skewed Student's (t) distributions. This allows the analysis to examine whether explicitly accounting for the non-normal characteristics commonly observed in financial returns improves volatility forecasting performance.

Under the normal specification, the standardized innovation follows a Gaussian distribution:

$$z_t \sim N(0,1) \tag{27}$$

with conditional density:

$$f(z_t) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z_t^2}{2}\right) \tag{28}$$

The Student's (t) specification introduces an additional degrees-of-freedom parameter, ν, allowing for heavier tails than the Gaussian distribution. The standardized Student's (t) density can be written as:

$$f(z;\nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)}} \left(1 + \frac{z^2}{\nu-2}\right)^{-\frac{\nu+1}{2}} \tag{29}$$

where $\Gamma(\cdot)$ denotes the gamma function. The degrees-of-freedom parameter controls the thickness of the distribution's tails, with lower values of ν corresponding to heavier tails.

Hansen's (1994) Skewed Student's (t)-distribution extends the standard Student's (t)-distribution by introducing a skewness parameter, λ, alongside the degrees-of-freedom parameter, ν. This allows the distribution to capture both asymmetric behavior and heavy tails commonly observed in financial return series. The probability density function of Hansen's (1994) Skewed Student's (t)-distribution can be expressed as follows:

$$f(z_t \mid \eta, \lambda) = \begin{cases} b \cdot c \left(1 + \frac{1}{\eta-2}\left(\frac{bz_t+a}{1-\lambda}\right)^2\right)^{-\frac{\eta+1}{2}} & \text{if } z_t < -\frac{a}{b} \\ b \cdot c \left(1 + \frac{1}{\eta-2}\left(\frac{bz_t+a}{1+\lambda}\right)^2\right)^{-\frac{\eta+1}{2}} & \text{if } z_t \geq -\frac{a}{b} \end{cases} \tag{30}$$

where $\eta > 2$ controls tail thickness, $\lambda \in (-1,1)$ controls asymmetry, and the scaling constants a, b, and c enforce the standardization constraints $\mathbb{E}[z_t] = 0$ and $\text{Var}(z_t) = 1$:

$$a = 4\lambda c \left(\frac{\eta - 2}{\eta - 1} \right), \quad b = \sqrt{1 + 3\lambda^2 - a^2}, \quad c = \frac{\Gamma\left(\frac{\eta+1}{2}\right)}{\sqrt{\pi(\eta - 2)}\,\Gamma\left(\frac{\eta}{2}\right)} \tag{31}$$

where the density is standardized through distribution-specific constants that depend on ν and λ. The skewed Student's (t) specification therefore allows the conditional innovation distribution to simultaneously accommodate excess kurtosis and asymmetry.

The normal distribution provides a conventional benchmark, whereas the Student's (t) distribution accommodates heavier tails and the skewed Student's (t) distribution additionally allows for asymmetry. Evaluating the three alternatives allows the distributional assumptions to be determined empirically rather than imposed a priori.

Cross-combining the three conditional variance specifications, three innovation distributions, and two conditional mean specifications results in 18 candidate specifications, defined as:

Table 5: GARCH-family Models Specification Space

Variance Dynamics (σ_t^2)	**Innovation Distribution** (z_t)	**Mean Specification** (μ_t)
GARCH(1,1)	Normal (Gaussian)	Constant (μ)
EGARCH(1,1)	Student's t	AR(1) ($\mu + \phi r_{t-1}$)
GJR-GARCH(1,1)	Skewed Student's t	

5.3.3 Estimation and Model Selection

The 18 candidate specifications are estimated using maximum likelihood-based methods. For each specification, the likelihood function is constructed according to the selected conditional mean, conditional variance, and innovation distribution, and the model parameters are obtained through numerical optimization.

Rather than selecting the final specifications solely on the basis of in-sample goodness-of-fit criteria, model selection is based primarily on out-of-sample forecasting performance. Each candidate model is estimated using the training data, and its forecasting performance is evaluated using the validation period. The specification that provides the best forecasting performance within each GARCH family is selected for subsequent out-of-sample analysis. This approach prioritizes the ability of the models to generalize to unseen observations rather than their in-sample fit.

The forecasts from all 18 candidate specifications are evaluated using QLIKE, MAE, and RMSE. QLIKE is used as the primary criterion for evaluating volatility forecasts, given its suitability for volatility forecast evaluation and its use as the loss function in the GARCH estimation. MAE and RMSE are reported as complementary measures of forecast accuracy. Lower values indicate better forecasting performance. In-sample information criteria such as

AIC and BIC are not used as model selection criteria, as the primary objective is to identify specifications with superior out-of-sample forecasting performance

The three models selected for the subsequent hybrid LSTM model consist of the best-performing specification from each GARCH family: one GARCH, one EGARCH, and one GJR-GARCH model. The selected specification within each family is determined based on out-of-sample forecasting performance across the alternative conditional mean and innovation-distribution specifications. Their one-day-ahead volatility forecasts are incorporated as additional input features, allowing the LSTM to learn from information provided by different volatility-modeling structures.

The single best-performing specification among the 18 candidate models is additionally selected as the GARCH component of the regime-aware ensemble model. Thus, the GARCH model used in the ensemble is determined empirically based on its out-of-sample forecasting performance, rather than being selected in advance according to its model class, conditional mean specification, or innovation distribution.

This model-selection procedure reduces the risk of selecting GARCH specifications solely on the basis of in-sample fit and ensures that the models passed to the subsequent hybrid and ensemble stages are supported by their out-of-sample forecasting performance. It also allows the empirical analysis to determine whether differences in conditional mean dynamics, volatility-model structure, or innovation distribution contribute to improved one-day-ahead volatility forecasts.

5.3.4 Rolling-Window Forecasting

The selected GARCH-family specifications are subsequently used to generate one-day-ahead volatility forecasts through a rolling-window procedure that preserves the temporal ordering of the observations. Several estimation window lengths are considered, and the final window length is chosen in line with the existing literature. The models are refitted every 22 trading days. Between refitting dates, the estimated parameters are kept fixed while newly available observations are incorporated into the rolling estimation window for subsequent forecasts. This procedure allows the models to incorporate evolving market information while limiting the computational burden associated with daily parameter re-estimation.

A fixed-length rolling-window framework is employed to estimate and forecast the GARCH-type models. Unlike an expanding-window approach, in which the estimation sample continuously increases over time, the rolling-window approach maintains a fixed number of the most recent observations for parameter estimation. This allows the model parameters to adapt to changes in volatility dynamics and evolving market conditions while limiting the influence of older observations that may no longer be representative of the current market environment.

Previous studies emphasize that the choice of estimation-window length can affect out-of-sample forecasting performance, particularly for financial time series with time-varying parameters and structural changes. Inoue, Jin, and Rossi (2017) treat the rolling-window length as a tuning parameter, while Foster and Nelson (1996) highlight the trade-off between responsiveness to changing market conditions and parameter estimation precision. Shorter windows adapt more quickly to recent changes but may yield less stable estimates, whereas longer windows provide greater estimation precision at the cost of reduced responsiveness. This trade-off is particularly relevant for asymmetric GARCH models, such as EGARCH and GJR-GARCH, which involve additional parameters to capture asymmetric responses to shocks. Awartani and Corradi (2005) emphasize the importance of accounting for such asymmetries in equity-market volatility modeling, suggesting that reliable identification of asymmetric effects requires sufficient information on the behavior of volatility following shocks. Accordingly, rolling-window selection should consider model complexity, parameter stability, and responsiveness, providing a rationale for using different window lengths across symmetric and asymmetric GARCH specifications. Therefore, the fixed-window framework is applied throughout the forecasting procedure, with the different window lengths used for each GARCH specification.

A 750-observation rolling window is adopted for the standard GARCH(1,1) model, providing sufficient observations for stable maximum likelihood estimation while allowing the model to incorporate recent volatility dynamics. This choice follows the empirical precedent of Opschoor, van Dijk, and van der Wel (2017) and Becker and Leschinski (2021), who also employ a 750-observation rolling window in their forecasting analyses. The approximately three-year estimation window is therefore considered appropriate for the relatively parsimonious symmetric GARCH specification, while longer windows are considered for the more parameter-intensive asymmetric specifications.

For the EGARCH and GJR-GARCH specifications, a longer rolling window of 1,000 trading days is used. The larger estimation sample is intended to provide greater information for estimating the additional parameters associated with asymmetric volatility dynamics. This specification is in line with the empirical settings employed by Kuester, Mittnik, and Paolella (2006) and Herwartz (2004), who use estimation windows of approximately 1,000 observations in their forecasting analyses. The approximately four-year window provides sufficient information for obtaining the reliable estimation of the additional parameters required to capture asymmetric volatility effects.

The 1,000-observation window is particularly appropriate for EGARCH and GJR-GARCH, as their asymmetric structures involve additional parameters that require a sufficiently large estimation sample for stable parameter estimates and reliable numerical convergence. This is especially relevant when these models are estimated with Student's t and skewed Student's t innovations, which introduce additional distributional parameters.

5.4 Standalone LSTM Model

5.4.1 Model specification

The second stage of the empirical analysis considers a standalone Long Short-Term Memory (LSTM) network as a non-parametric deep learning approach to Nasdaq-100 realized volatility forecasting. The LSTM is selected because its recurrent architecture is designed to capture temporal dependencies and nonlinear relationships in sequential data.

The dependent variable is the one-day-ahead realized volatility of the Nasdaq-100 index. The explanatory variables consist of transformed lagged volatility information available at the time of forecasting. In particular, the model incorporates historical realized volatility measures over different horizons. All explanatory variables are lagged appropriately to ensure that only information available prior to the forecast date is used.

An LSTM network consists of a sequence-processing layer followed by an output layer. At each time step, the network receives an input vector x_t and maintains a hidden state h_t and a cell state c_t. The cell state allows information to be retained over longer sequences, while gating mechanisms determine which information should be added, retained, or removed.

The main LSTM operations can be represented as

- **Forget Gate:** Decides what old information to throw away

$$f_t = \sigma\big(W_f \cdot [h_{t-1}, x_t] + b_f\big) \tag{32}$$

- **Input Gate:** Decides what new information to add

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{33}$$

- **Candidate Cell State:** Creates new candidate values

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{34}$$

- **Cell State Update:** Combines old and new memory

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{35}$$

- **Output Gate:** Decides what part of the cell state to output

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{36}$$

- **Hidden State (Output):** Final output for the current step

$$h_t = o_t \odot \tanh(c_t) \tag{37}$$

where f_t, i_t, and o_t denote the forget, input, and output gates, respectively. The vectors c_t and h_t represent the cell and hidden states, while W and b denote the corresponding weight matrices and biases.

The LSTM therefore provides a flexible nonlinear mapping between the historical input sequence and future realized volatility without imposing a predefined parametric relationship.

5.4.2 Input Variables and Sequence Construction

The LSTM model is designed to process the input data as sequential observations rather than as independent observations. For each forecasting date t, a fixed-length sequence of past observations is provided to the network, which then generates a one-day-ahead forecast of realized volatility, RV_{t+1}. A 22-day lookback window is used to define the length of each input sequence, with the model using the observations from the preceding 22 trading days to generate the forecast.

The input variables include rolling realized volatility measures over short- and medium-term horizons described in Chapter 3. The use of multiple volatility horizons is motivated by the persistent nature of financial volatility and the possibility that recent and longer-term volatility conditions contain different information about future volatility. Specifically, $RV_{1d,t}$, $RV_{5d,t}$, $RV_{22d,t}$ were initially considered to represent daily, weekly, and approximately monthly volatility information, respectively.

The input variables are standardized before model training. Scaling parameters are estimated exclusively from the training data and subsequently applied to the validation and test observations, as described in Section 3.4. The standardization is performed using the following transformation:

$$x_t^{\text{scaled}} = \frac{x_t - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{38}$$

where x_t denotes the original observation, while μ_{train} and σ_{train} represent the mean and standard deviation estimated from the training data, respectively. The resulting transformation is then applied to the corresponding observations in the validation and test periods using these fixed training-period parameters.

5.4.3 Training and hyperparameter optimization

Unlike the GARCH-family models, which are estimated through maximum likelihood, the LSTM parameters are obtained through numerical optimization of a specified loss function. The model is trained by minimizing the Mean Absolute Error (MAE) between the predicted and observed realized volatility. MAE is selected as the training loss because it measures the absolute deviation between forecasts and the realized volatility proxy while being less sensitive to extreme observations than squared-error-based losses. This is particularly relevant for the Nasdaq-100 volatility data examined in the descriptive and diagnostic analysis, which exhibit pronounced volatility spikes during periods of market stress. Using MAE therefore helps prevent a small number of extreme volatility observations from disproportionately influencing the parameter optimization, allowing the model to focus on reducing the typical magnitude of forecasting errors.

A walk-forward forecasting procedure is employed to preserve the time-series structure of the data. An initial training window of 1,500 observations is used to provide a sufficiently large estimation sample, which helps stabilize the model before forecasting begins. A 126-observation validation window is employed for model validation, and the forecasting procedure advances in 126-observation steps. At each forecasting point, the model is estimated using only information available up to that point, and the subsequent observation is forecast. This procedure preserves the temporal ordering of the data, prevents look-ahead bias, and provides a realistic assessment of out-of-sample forecasting performance.

The LSTM models are trained using the Adam gradient-based optimization algorithm, with the learning rate controlling the magnitude of parameter updates. Hyperparameters are selected using Optuna based on validation-set performance. The search considers the number of LSTM layers, number of hidden units, dropout rate applied to the outputs of the LSTM layers, recurrent dropout applied to the recurrent connections within the LSTM units, and learning rate. During hyperparameter optimization, training is limited to a maximum of 10 epochs, with early stopping applied using a patience of 5 epochs to reduce the risk of overfitting. The hyperparameter configuration that achieves the best validation performance is retained for the subsequent model estimation.

Following hyperparameter selection, the LSTM model is re-estimated using the combined training and validation sets, while keeping the selected hyperparameter configuration fixed. To accommodate the larger training sample, the maximum number of epochs is increased to 50, with early stopping applied using a patience of 7 epochs based on the training loss. The refitted model then generates the one-step-ahead forecasts for the out-of-sample test period, ensuring that all observations available before the test period are used for final model estimation.

5.5 Hybrid GARCH-LSTM Model

The first proposed integration strategy combines the information contained in several GARCH forecasts with the nonlinear learning capabilities of the LSTM.

Let $\hat{RV}_{t+1}^{(1)}, \quad \hat{RV}_{t+1}^{(2)}, \quad \hat{RV}_{t+1}^{(3)}$ denote the one-day-ahead volatility forecasts generated by the three best-performing GARCH-family models. These forecasts are incorporated into the LSTM input vector together with the other selected explanatory variables.

The hybrid model can therefore be represented conceptually as:

$$RV_{t+1} = f_{\text{LSTM}}\left(X_{t+1}, \hat{RV}^{(1)}_{t+1}, \hat{RV}^{(2)}_{t+1}, \hat{RV}^{(3)}_{t+1}\right) \tag{39}$$

where $X_{t+1} = [RV_{1d,t+1}, RV_{22d,t+1}]$ represents the lagged realized volatility inputs available at time t+1, and f_{LSTM} denotes the nonlinear mapping learned by the LSTM network.

This represents a feature-level integration strategy. Instead of directly averaging the GARCH forecasts with the LSTM forecast, the network is allowed to learn how information from the different GARCH specifications should contribute to the final prediction.

The hybrid LSTM model uses the same hyperparameter configuration as the standalone LSTM model, including the network architecture, dropout settings, learning rate, batch size, and training parameters. This ensures that the comparison between the standalone and hybrid specifications reflects the effect of incorporating GARCH-based forecasts as additional input features rather than differences in hyperparameter tuning. All candidate features are examined using a correlation matrix to identify substantial redundancy among the inputs and assess their relationships with the target variable. The full correlation matrix is presented in Appendix C.2.

5.6 Gaussian Mixture Model for Market Regime Identification

The second integration strategy incorporates information about market conditions through a Gaussian Mixture Model. The purpose of the GMM is to identify latent market regimes characterized by different volatility conditions.

For an observation x_t, the probability density under a K-component Gaussian mixture is

$$\sum_{k=1}^{K} \pi_k \mathcal{N}(x_t \mid \mu_k, \Sigma_k) \tag{40}$$

where π_k denotes the mixing probability of regime k, μ_k and Σ_k represent the mean and covariance parameters of that regime, and the mixing probabilities satisfy

$$\sum_{k=1}^{K} \pi_k = 1 \tag{41}$$

Rather than assigning observations deterministically to a single cluster, the GMM produces posterior probabilities of belonging to each regime:

$$\frac{\pi_k \mathcal{N}(x_t \mid \mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_t \mid \mu_j, \Sigma_j)} \tag{42}$$

The resulting probabilities are interpreted as measures of the likelihood that the market belongs to each identified volatility regime. The regimes are subsequently characterized as relatively calm or crisis conditions based on their estimated volatility characteristics.

Several feature configurations are evaluated for market regime identification, starting with nine candidate features and progressively reducing the number of features. Before applying the GMM, all candidate features are standardized to ensure that differences in scale do not disproportionately influence the clustering results. At each stage, boxplots on the validation set are examined to assess the separation between the identified regimes, and features showing substantial overlap are removed. For each number of retained features, alternative feature combinations are compared using the Bayesian Information Criterion (BIC), calculated as:

$$\mathrm{BIC} = p \ln(T_{\mathrm{obs}}) - 2 \ln \mathscr{L}(\hat{\theta}) \tag{43}$$

where p is the total number of estimated parameters, T_{obs} is the total number of in-sample observations, and $\ln \mathscr{L}(\hat{\theta})$ is the maximized total sample log-likelihood, defined as:

$$\ln \mathscr{L}(\hat{\theta}) = \sum_{t=1}^{T_{\mathrm{obs}}} \ln p(x_t \mid \hat{\theta}) \tag{44}$$

The combination with the lowest BIC is selected. This process is repeated as the number of features decreases. The best model selected at each feature-count level is then compared using the mean log-likelihood $\bar{\ell}(\hat{\theta})$, calculated as:

$$\bar{\ell}(\hat{\theta}) = \frac{1}{T_{\mathrm{obs}}} \sum_{t=1}^{T_{\mathrm{obs}}} \ln p(x_t \mid \hat{\theta}) = \frac{\ln \mathscr{L}(\hat{\theta})}{T_{\mathrm{obs}}} \tag{45}$$

where $p(x_t \mid \hat{\theta})$ is the probability density of observation x_t under the GMM, and $\hat{\theta}$ denotes the estimated model parameters. The model with the highest mean log-likelihood value $\bar{\ell}(\hat{\theta})$ is selected as the final GMM specification.

BIC is used during the within-stage selection because it accounts for both model fit and model complexity, whereas mean log-likelihood is used for the final comparison of the best specifications across different feature counts. Unlike BIC, it does not impose an additional penalty for model complexity, allowing the final comparison to focus on the ability of the selected models to represent the observed data.

The GMM features are selected to capture different aspects of recent volatility dynamics and market uncertainty. The first feature is the exponential moving average (EMA) of realized volatility, defined as follows:

$$\mathrm{EMA}^{\mathrm{lag}}_{m,t} = \alpha_m RV_{t-1} + (1 - \alpha_m)\,\mathrm{EMA}^{\mathrm{lag}}_{m,t-1}, \qquad \alpha_m = \frac{2}{m+1} \tag{46}$$

with m=5 and m=22, corresponding to EMA_5 and EMA_{22}, respectively.

The rolling standard deviation of returns is calculated as:

$$\sigma^{\mathrm{lag}}_{w,t} = \frac{1}{w-1} \sum_{i=1}^{w} \left(r_{\mathrm{close},t-i} - \bar{r}_{w,t-1} \right)^2 \tag{47}$$

where $\bar{r}_{w,t}$ denotes the mean return over the corresponding w-day window. The values

w=5 and w=22 are used to obtain σ_{5d} and σ_{22d}.

The Parkinson volatility feature is constructed from the daily high and low prices. The log-transformed Parkinson variance measure is defined as:

$$\ln(\sigma_{\text{Parkinson},t}) = \ln\sqrt{\frac{1}{4\ln(2)}\left[\ln\left(\frac{H_{t-1}}{L_{t-1}}\right)\right]^2} \tag{48}$$

Finally, the VXN variable is transformed logarithmically as $ln(VXN_t)$ to reduce scale-related effects and account for the typically right-skewed distribution of the volatility index.

5.7 Regime-Aware Ensemble Model

The second proposed approach combines the forecasts of the best-performing GARCH model and the standalone LSTM model. Unlike the hybrid LSTM, which integrates GARCH forecasts at the feature level, this approach performs integration at the forecast level.

Building on the motivation discussed in the literature review, a regime-aware ensemble is constructed to allow the relative contribution of the GARCH and LSTM forecasts to vary according to prevailing market conditions. Market regimes are identified using a Gaussian Mixture Model (GMM), which provides a probability-based classification of each observation into the calm and crisis regimes. Rather than assigning observations deterministically to a single regime, the estimated regime probabilities are used to determine the time-varying ensemble weights.

Let $R\hat{V}_{t+1}^{GARCH}$ denote the forecast from the best-performing GARCH model and $R\hat{V}_{t+1}^{LSTM}$ denote the standalone LSTM forecast. The ensemble forecast is expressed as:

$$R\hat{V}_{t+1}^{\text{ENS}} = \pi_t R\hat{V}_{t+1}^{\text{GARCH}} + (1-\pi_t) R\hat{V}_{t+1}^{\text{LSTM}} \tag{49}$$

where π_t represents the time-varying weight assigned to the GARCH forecast and $1-\pi_t$ represents the corresponding weight assigned to the LSTM forecast. The weight π_t is determined by the estimated probability of the calm regime, such that a higher probability of a calm market results in a greater weight assigned to the GARCH forecast. Conversely, as the probability of the crisis regime increases, a greater weight is assigned to the LSTM forecast. This allows the ensemble to adjust continuously to changes in the estimated market regime rather than relying on a fixed weighting scheme.

5.8 Forecast Evaluation and Statistical Comparison

The forecasting performance of all models is evaluated using the same out-of-sample observations. Three principal evaluation measures are used: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and QLIKE.

MAE is defined as:

$$MAE = \frac{1}{T}\sum_{t=1}^{T} |RV_t - \hat{RV}_t| \tag{50}$$

RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}^{T} \left(RV_t - \hat{RV}_t\right)^2} \tag{51}$$

QLIKE is calculated as:

$$\text{QLIKE} = \frac{1}{T}\sum_{t=1}^{T} \left(\ln(\hat{RV}_t^2) + \frac{RV_t^2}{\hat{RV}_t^2} \right) \tag{52}$$

Lower values indicate better forecasting performance for all three measures. QLIKE is specifically designed for evaluating volatility and variance forecasts, as it accounts for the asymmetric consequences of under- and over-predicting the conditional variance. MAE provides an interpretable measure of average forecast error and is relatively less sensitive to extreme observations. In contrast, RMSE places greater weight on large forecast errors because the errors are squared, making it more sensitive to outliers and particularly informative about forecast performance during periods of substantial volatility. The use of multiple evaluation metrics provides a more comprehensive assessment of model performance across different error characteristics.

To determine whether differences in forecasting performance are statistically significant, the Diebold-Mariano (DM) test is applied to competing forecast loss series. For two competing models, the loss differential is defined as:

$$\Delta_t = \mathscr{L}_{\text{loss}}(e_{1,t}) - \mathscr{L}_{\text{loss}}(e_{2,t}) \tag{53}$$

where $e_{i,t}$ denotes the forecast error of model i at time t. The function $\mathscr{L}_{\text{loss}}$ denotes the selected loss function. Under the null hypothesis, the two models have equal predictive accuracy. A statistically significant result provides evidence that the forecasting performance of the two models differs.

Since the true conditional volatility is unobservable, forecast evaluation relies on the Yang–Zhang realized volatility proxy, which is subject to measurement error.Patton (2011) demonstrates that forecast evaluation can be sensitive to the choice of loss function when the true volatility is unobservable and an imperfect proxy is used, as measurement noise can affect the relative ranking of competing forecasts. QLIKE is among the class of loss functions whose forecast rankings are robust to such noise, making it particularly suitable for comparing volatility forecasts in this setting. Accordingly, QLIKE is adopted as the primary loss function for the Diebold–Mariano tests, with the same loss function applied to all competing out-of-sample forecasts.

5.9 Methodological Assumptions, Properties, and Limitations

The empirical framework developed in this study exhibits distinct structural properties and inherent limitations across the econometric, deep learning, and regime-switching methodologies.

The classic econometric models—comprising standard GARCH, GJR-GARCH, and EGARCH specifications—excel at capturing volatility clustering and asymmetric leverage dynamics. Econometrically, these models provide fully transparent parameterizations and asymptotic inference. However, these parametric specifications suffer from significant limitations, including an inability to adapt rapidly to structural breaks, reliance on restrictive distributional assumptions, and short-memory exponential decay that can underestimate persistent high-volatility regimes.

The standalone LSTM framework leverages recurrent memory cells governed by input, forget, and output gates to learn complex non-linear temporal dependencies across lookback windows. By mitigating vanishing gradient problems, the architecture effectively extracts persistent non-linear patterns from historical volatility sequences that traditional linear time series models fail to detect. Conversely, this standalone approach suffers from a total lack of parameter interpretability, operating purely as a black-box estimator. Furthermore, its vast parameter space renders the network highly sensitive to financial market noise and parameter estimation instability, requiring rigorous regularization techniques—such as dropout and early stopping—to maintain out-of-sample generalization.

The Hybrid LSTM architecture expands upon this foundation by integrating structural econometric signals alongside raw sequential inputs to guide the deep learning optimization process. By fusing parametric volatility estimations with non-parametric recurrent layers, the hybrid formulation aims to bridge empirical domain knowledge with non-linear feature extraction, constraining the search space against arbitrary financial noise. This hybrid design provides flexibility but may also introduce redundant or highly correlated information because forecasts from different GARCH specifications can exhibit similar dynamics. Additionally, calibrating the joint feature space increases computational overhead and optimization difficulty, making hyperparameter tuning far more delicate than in standard recurrent models.

The Gaussian Mixture Model (GMM) assumes that the observed data can be represented as a finite mixture of Gaussian distributions, with each component representing a distinct statistical regime. Although GMM provides a flexible probabilistic approach to regime identification, parameter estimation via the Expectation-Maximization algorithm remains vulnerable to local log-likelihood optima and exhibits high sensitivity to initial candidate feature selection. Unlike Markov-switching models, GMM does not explicitly account for the time dependence or transition probabilities between regimes. Instead, it identifies regime membership mainly from the observed characteristics of the data rather than from how the underlying states change over time. Consequently, the absence of explicit transition dynamics may limit the ability of GMM to accurately forecast future regime probabilities, which may affect its usefulness in regime-aware volatility forecasting.

Finally, none of the considered approaches can be assumed to be universally optimal. GARCH-family models offer interpretability and a well-established econometric framework but impose a parametric structure on volatility dynamics. LSTM models provide greater flexibility in capturing nonlinear temporal relationships but are less interpretable and more sensitive to model specification. The hybrid and ensemble approaches introduce additional sources of information but also increase methodological complexity and estimation uncertainty. The relative merits of these approaches are therefore determined empirically through their out-of-sample forecasting performance and statistical comparison rather than through theoretical assumptions about which model should perform best.

6 ESTIMATION RESULTS

6.1 GARCH-Family Model Results

6.1.1 Forecasting performance

Convergence diagnostics were performed for all estimated model specifications to ensure that the optimization procedures converged successfully and that the resulting parameter estimates were reliable. All estimated models met the required convergence conditions, confirming the validity of the estimation results used in the subsequent forecasting analysis.

Table 6: Model Performance Metrics (GARCH and ARMA-GARCH)

Model	MAE	RMSE	QLIKE
ARMA-GARCH_skewt	0.2241	0.3101	1.3180
GARCH_skewt	0.2273	0.3175	1.3185
GARCH_normal	0.2362	0.3281	1.3200
ARMA-GARCH_normal	0.2403	0.3307	1.3204
GARCH_t	0.2437	0.3404	1.3206
ARMA-GARCH_t	0.2460	0.3422	1.3206

Table 6 shows that the ARMA-GARCH model with a skewed Student's t innovation distribution achieves the best overall forecasting performance among the GARCH-type specifications, recording the lowest QLIKE, MAE, and RMSE values.

Table 7: Model Performance Metrics (EGARCH and ARMA-EGARCH)

Model	MAE	RMSE	QLIKE
EGARCH_normal	0.2429	0.3334	1.3210
ARMA-EGARCH_normal	0.2494	0.3396	1.3217
ARMA-EGARCH_skewt	0.2794	0.3890	1.3250
EGARCH_skewt	0.2834	0.3947	1.3257
EGARCH_t	0.2871	0.3998	1.3262
ARMA-EGARCH_t	0.2890	0.4024	1.3265

Table 7 indicates that the EGARCH model with a normal innovation distribution performs best among the EGARCH specifications, achieving the lowest values across the three evaluation metrics.

Similarly, Table 8 shows that the GJR-GARCH model with a normal innovation distribution is the best-performing specification within the GJR-GARCH family.

Based on these results, the best-performing specification from each GARCH family is selected for the subsequent hybrid LSTM model. Accordingly, three models are carried forward: ARMA-GARCH with a skewed Student's t innovation distribution, EGARCH with a normal

Table 8: Model Performance Metrics (GJR-GARCH and ARMA-GJR-GARCH)

Model	MAE	RMSE	QLIKE
GJR_normal	0.2553	0.3572	1.3228
ARMA-GJR_normal	0.2585	0.3577	1.3230
ARMA-GJR_skewt	0.2810	0.3914	1.3257
ARMA-GJR_t	0.2945	0.4119	1.3275
GJR_skewt	0.2939	0.4111	1.3279
GJR_t	0.3003	0.4218	1.3287

innovation distribution, and GJR-GARCH with a normal innovation distribution. Collectively, the results show that normally distributed errors provide the strongest predictive performance for the asymmetric GARCH models, suggesting that their richer conditional variance dynamics may already capture the key features of the volatility process. In contrast, the standard GARCH model benefits from more flexible error distributions, with the skewed Student's t specifications providing improved forecasting performance.

6.1.2 Selected GARCH-family specifications

Table 9: Performance evaluation metrics comparison across top models

Model	QLIKE
ARMA-GARCH_skewt	1.3180
EGARCH_normal	1.3210
GJR-GARCH_normal	1.3228

In summary, Table 9 compares the forecasting performance of the three best-performing specifications from each GARCH family. Based on the QLIKE criterion, ARMA-GARCH with a skewed Student's t innovation distribution achieves the best overall performance and is therefore selected for the subsequent ensemble model.

6.1.3 Visual comparison of volatility forecasts

To complement the quantitative evaluation presented above, the one-day-ahead volatility forecasts of the three selected GARCH-family models are visually compared with the realized volatility. This comparison provides an intuitive assessment of how well the models capture the dynamics and fluctuations of realized volatility over the out-of-sample period.

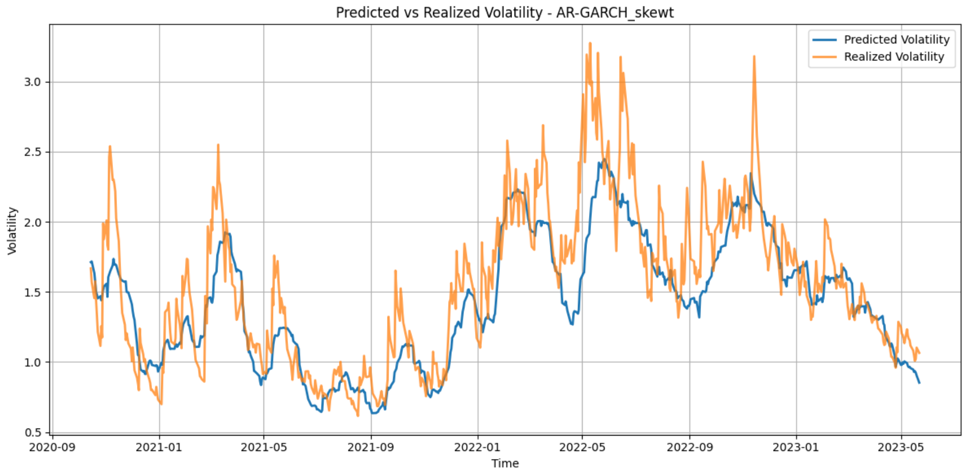

Figure 5: Predicted and Realized Volatility - ARMA-GARCH_skewt.

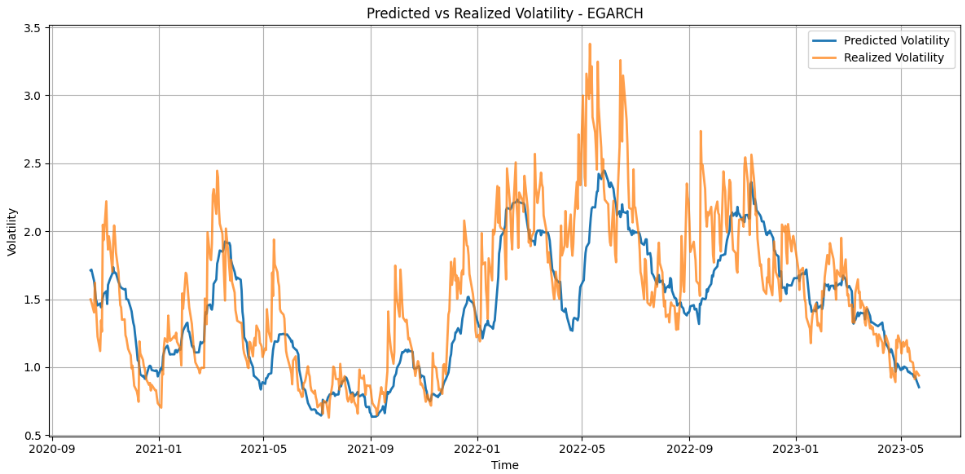

Figure 6: Predicted and Realized Volatility - EGARCH.

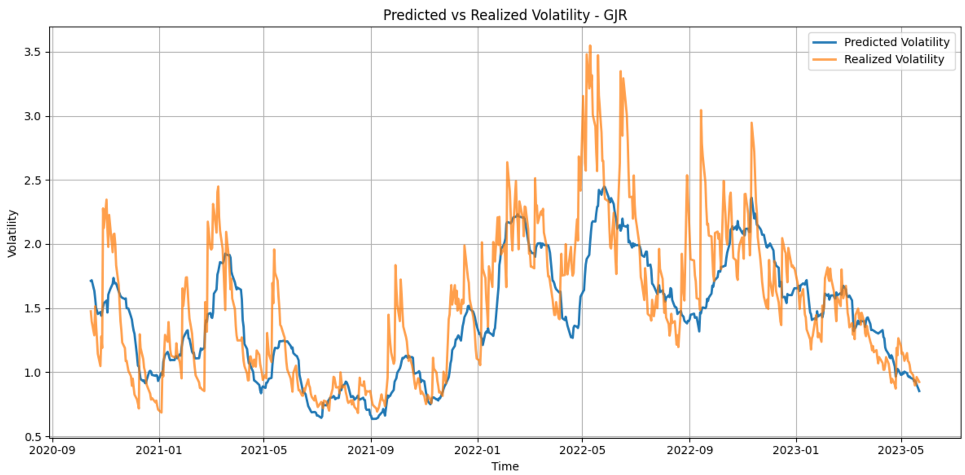

Figure 7: Predicted and Realized Volatility - GJR-GARCH.

Figures 5, 6, and 7 present the volatility forecasts generated by the three selected GARCH-family models against the realized volatility. Overall, the three models exhibit broadly similar forecasting patterns. Their forecasts are relatively smooth and tend to underestimate sharp short-term fluctuations and sudden volatility spikes. Although all three models capture the broader upward and downward movements in volatility, they respond less rapidly to abrupt changes, with major volatility peaks often being captured with a lag. Given the strong visual similarity among the forecasts, the quantitative evaluation metrics provide a more informative basis for distinguishing their relative forecasting performance.

6.1.4 In-sample model diagnostics

To assess the adequacy and reliability of the selected GARCH-family specifications beyond their out-of-sample forecasting performance, several in-sample model diagnostics are conducted. For this analysis, a single static model specification is estimated for each selected GARCH-family model using the full in-sample period, rather than applying the rolling estimation procedure used for out-of-sample forecasting. These diagnostics examine whether the models adequately capture the conditional mean and volatility dynamics and whether the standardized residuals exhibit any remaining systematic dependence or heteroskedasticity. The results provide additional evidence on the overall specification quality of the selected models.

Table 10: Ljung-Box test results on standardized residuals across model specifications

	ARMA-GARCH_skewt		EGARCH		GJR-GARCH	
Lag	Statistic	p-value	Statistic	p-value	Statistic	p-value
10	18.7001	0.0442	22.4285	0.0131	12.4932	0.0344
15	24.9618	0.0505	29.2210	0.0151	26.2327	0.0356
20	32.9052	0.0346	39.5282	0.0057	34.7933	0.0212

At the 1% significance level ($\alpha = 0.01$), the standardized residuals of the three selected models are generally consistent with white noise across the tested lag lengths, indicating that the models adequately capture the serial dependence in the return series. However, the EGARCH model at 20 lags produces a p-value of 0.0057, leading to rejection of the null hypothesis of no residual autocorrelation at the 1% significance level. This suggests that some residual serial dependence may remain in the EGARCH specification at this lag length. Nevertheless, the overall results indicate that the selected models largely capture the linear dependence in the return series.

At the 1% significance level and across the tested lag lengths, the Ljung–Box test applied to the squared standardized residuals indicates that all three models sufficiently capture the conditional heteroskedasticity in the return series. The test results provide no significant evidence of remaining autocorrelation in the squared standardized residuals, suggesting that the selected GARCH-family specifications successfully account for the volatility dynamics in the data.

Table 11: Ljung-Box test results on squared standardized residuals across model specifications

	ARMA-GARCH_skewt		EGARCH		GJR-GARCH	
Lag	Statistic	p-value	Statistic	p-value	Statistic	p-value
10	12.3106	0.2648	12.4932	0.2534	12.2689	0.2675
15	22.6182	0.0926	16.1965	0.3691	17.8194	0.2723
20	33.3719	0.0307	35.1898	0.0191	24.9159	0.2047

To further support this conclusion, the ARCH-LM test is applied to the standardized residuals of the three selected models, with the results presented in Table 12. The high p-values associated with both the LM and F-statistics indicate that the null hypothesis of no remaining ARCH effects cannot be rejected. This provides further evidence that the selected models appropriately capture the conditional heteroskedasticity in the return series, with no significant ARCH effects remaining in the standardized residuals.

To examine the standardized residuals visually, Q-Q plots and ACF plots are presented in Appendices A.1, A.2, and A.3 for each model. The ACF plots indicate no significant autocorrelation in the standardized residuals, suggesting that the models adequately capture the serial dependence in returns. However, the Q-Q plots indicate deviations from the assumed residual distributions, particularly in the tails, suggesting that the distributional assumptions are not fully satisfied.

Table 12: ARCH effect test results on standardized residuals across model specifications

Model	LM Statistic	LM p-value	F-statistic	F p-value
ARMA-GARCH_skewt	4.362	0.499	0.872	0.499
EGARCH	5.854	0.321	1.171	0.321
GJR-GARCH	5.940	0.312	1.188	0.312

Table 13: Descriptive statistics of standardized residuals across model specifications

Model	Mean	Std. Dev.	Skewness	Kurtosis
ARMA-GARCH_skewt	-0.0430	1.0041	-0.4083	1.3623
EGARCH	-0.0072	1.0003	-0.4683	1.4205
GJR-GARCH	-0.0090	1.0002	-0.4191	1.2431

Table 13 shows that, the standardized residuals from all three models have means that are close to zero, with the EGARCH and GJR-GARCH specifications showing particularly small deviations from zero. Their standard deviations are also approximately equal to one, suggesting that the estimated conditional volatilities provide a reasonable scaling of the return innovations. Overall, these descriptive statistics indicate that the residuals are broadly consistent with the expected properties of standardized innovations, although further diagnostic tests are required to assess the adequacy of the model specifications.

Beyond their first two moments, the standardized residuals exhibit clear departures from normality. As shown in Table 13, the residuals from the EGARCH and GJR-GARCH models display negative skewness and elevated positive kurtosis, indicating asymmetry and heavier tails than those of a normal distribution. To formally assess this departure, the Jarque–Bera test was conducted. As presented in Table 14, the p-values for both models are very small, leading to rejection of the null hypothesis of normality. These results confirm that the standardized residuals retain significant non-normal and fat-tailed characteristics.

For the ARMA-GARCH model with a skewed Student's t innovation distribution, a two-sample Kolmogorov–Smirnov (KS) test was conducted to assess the consistency between the empirical distribution of the standardized residuals and the fitted skewed Student's t distribution using the same estimated parameters, degrees-of-freedom parameter and skewness parameter. The test produces a small p-value, leading to rejection of the null hypothesis that the two distributions are identical. This result indicates a statistically significant difference between the empirical residual distribution and the fitted skewed Student's t distribution, suggesting that the assumed innovation distribution does not fully capture the distributional characteristics of the standardized residuals.

Table 14: Residual distribution diagnostic tests across model specifications

Model	Diagnostic Test	p-value
ARMA-GARCH_skewt	KS Stat	0.0012
EGARCH	JB Test	< 0.0001
GJR-GARCH	JB Test	< 0.0001

As the ARMA-GARCH model with skewed Student's t innovations achieves the best overall out-of-sample forecasting performance, an additional parameter stability analysis is conducted to examine whether its volatility dynamics remain reasonably stable over time. The model is re-estimated using a rolling-window approach, with 95% confidence intervals reported for three key parameters: volatility persistence ($\alpha + \beta$), the degrees of freedom parameter (η), and the skewness parameter (λ). The corresponding rolling-parameter plots are provided in Appendix B. Overall, the persistence parameter remains consistently close to unity throughout the sample, indicating that volatility shocks are highly persistent and decay only slowly over time. The degrees of freedom parameter exhibits considerable variation, ranging from approximately 6 to above 15, suggesting substantial changes in the tail behavior of returns across different periods. Lower values of η indicate more pronounced heavy-tailed behavior, whereas higher values correspond to relatively thinner tails. The skewness parameter also remains predominantly negative across the rolling windows, indicating persistent asymmetry in the conditional return distribution. Taken together, these results provide empirical support for allowing both time-varying tail thickness and asymmetry in the innovation distribution, consistent with the use of the skewed Student's t specification.

The substantial variation in the rolling parameter estimates indicates that the underlying volatility dynamics and distributional characteristics of NDX returns evolve over time. While the rolling GARCH framework allows the model parameters to adapt to changing market conditions, the observed variation also suggests that the relationship between past information and future volatility may be complex and potentially nonlinear. In addition, the remaining model misspecification suggests that the GARCH specifications may not fully capture all relevant patterns in the return dynamics. This provides motivation for examining whether LSTM models can identify additional temporal patterns beyond those represented by the parametric GARCH structure, motivating the subsequent comparison of standalone and hybrid forecasting approaches.

6.2 Standalone LSTM Results

The correlation analysis was first conducted among the initial candidate variables to assess potential redundancy. A very high correlation was observed between $RV_{1d,t}$ and $RV_{5d,t}$ ($\rho = 0.99$), indicating substantial overlap in the information captured by these two variables. The full correlation matrix is presented in Appendix C.1. Based on this result, $RV_{5d,t}$ was excluded, while $RV_{1d,t}$ and $RV_{22d,t}$ were retained as the final realized volatility inputs.

Table 15: Optuna categorical hyperparameter search space and optimal values for the standalone LSTM architecture

Hyperparameter	Search Space / Range	Selected Value
Number of Layers	$\{1,2\}$	1
Hidden Units	$\{32,64,128,256\}$	128
Dropout Rate	$\{0.1,0.2,0.3,0.4\}$	0.2
Recurrent Dropout	$\{0.0,0.1,0.2\}$	0.1
Learning Rate	$\{0.0001,0.0005,0.001,0.005,0.01,0.05\}$	0.01

Table 15 presents the hyperparameter search space and the final values selected for the standalone LSTM model. Dropout and recurrent dropout are incorporated as regularization techniques to reduce the risk of overfitting. The training and test-set performance of the resulting model is reported in Table 16. The MAE and RMSE values are relatively similar across the training and test sets, indicating that the model generalizes reasonably well to unseen observations. Although a larger difference is observed in QLIKE, this metric is more sensitive to the relative magnitude of forecast errors, particularly when realized volatility is low, and therefore may exhibit greater variation between samples. Overall, the results do not indicate substantial evidence of overfitting in the standalone LSTM model.

Table 16: In-sample and out-of-sample forecasting performance metrics - Standalone LSTM

Sample	RMSE	MAE	QLIKE
Train	0.1026	0.0560	1.4837
Test	0.0994	0.0569	1.0754

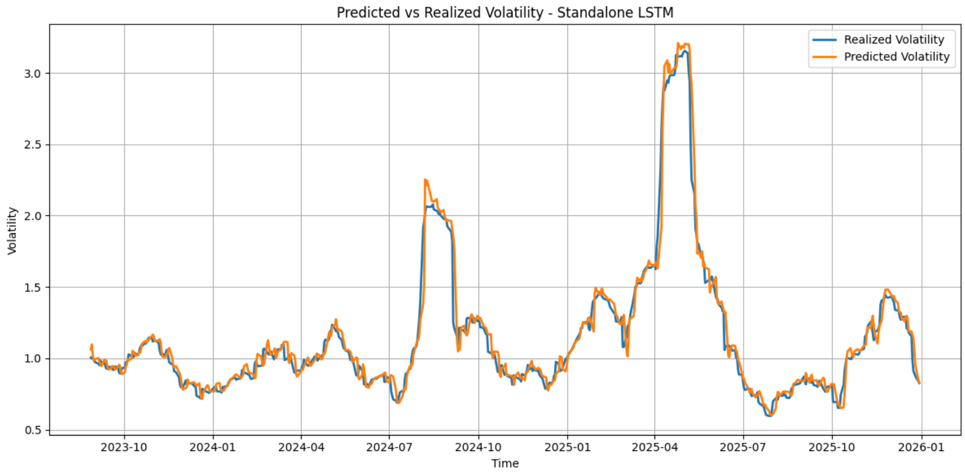

Figure 8: Predicted and Realized Volatility - Standalone LSTM.

Figure 8 shows that the standalone LSTM closely tracks the realized volatility throughout the out-of-sample period, capturing both its short-term fluctuations and major volatility peaks. In particular, during the two most pronounced periods of high-volatility conditions, the model responds promptly to the sharp changes in realized volatility, although it slightly overestimates the magnitude of these peaks. Overall, the visual comparison suggests that the standalone LSTM is capable of capturing both the persistent and rapidly changing patterns in realized volatility.

6.3 Hybrid LSTM Results

The correlation analysis shows that all selected features exhibit a relatively strong correlation with the target variable, with correlation coefficients exceeding 0.66. The highest pairwise correlation among the individual features is 0.96. Based on these results, there is insufficient evidence to exclude any of the selected features, as each provides relevant information for the prediction task.

The hybrid LSTM uses a higher recurrent dropout rate and a lower learning rate compared with the standalone LSTM. These hyperparameters were selected through the optimization process and may be related to the increased model complexity arising from the additional GARCH-based inputs. As shown by the performance metrics in Table 18, the model achieves relatively comparable results on the training and test sets, with no substantial deterioration in out-of-sample performance. This indicates that the hybrid model maintains stable forecasting

Table 17: Optuna categorical hyperparameter search space and optimal values for the hybrid architecture

Hyperparameter	Search Space / Range	Selected Value
Number of Layers	$\{1,2\}$	1
Hidden Units	$\{32,64,128,256\}$	128
Dropout Rate	$\{0.1,0.2,0.3,0.4\}$	0.2
Recurrent Dropout	$\{0.0,0.1,0.2\}$	0.2
Learning Rate	$\{0.0001,0.0005,0.001,0.005,0.01,0.05\}$	0.005

performance when applied to unseen data and does not exhibit pronounced overfitting.

Table 18: In-sample and out-of-sample forecasting performance metrics - Hybrid LSTM

Sample	RMSE	MAE	QLIKE
Train	0.1217	0.0586	1.3130
Test	0.0829	0.0506	1.0724

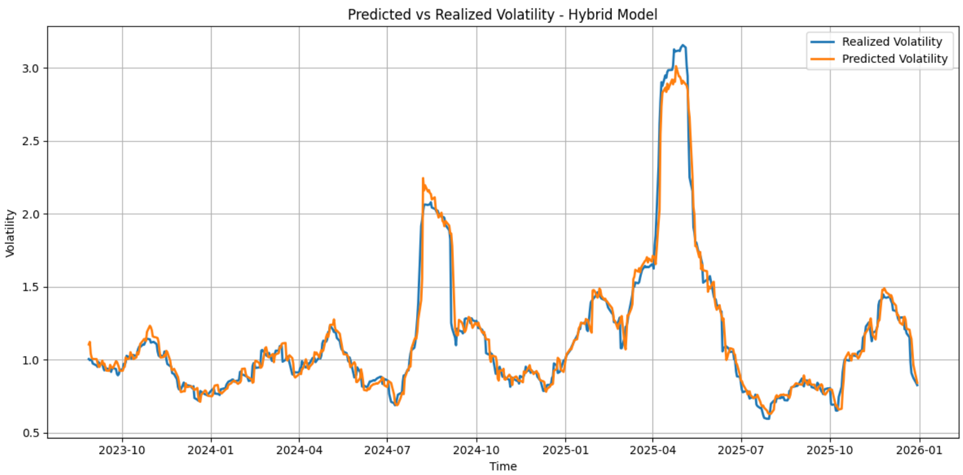

Figure 9: Predicted and Realized Volatility - Hybrid Model.

Figure 9 illustrates that the hybrid model's ability to capture volatility spikes varies across the sample. While it slightly overestimates realized volatility during the first major peak, it fails to fully capture the magnitude of the second peak, resulting in an underestimation. Despite capturing the general direction of volatility movements in several intervals, the forecasts tend to respond with a delay to abrupt changes, reflecting a similar lagging behavior to that observed in the GARCH-family models.

6.4 GMM Regime Identification Results

The first step in the GMM regime identification analysis is to determine the feature set that provides the best validation-set performance. Different combinations of candidate variables are evaluated. The preferred specification is selected based on its ability to distinguish the underlying volatility regimes while providing a good balance between model fit and parsimony.

Table 19: Binary feature inclusion matrix across GMM selection iterations

Number of features	RV_{1d}	RV_{5d}	RV_{22d}	$\mathbf{EMA}_5$	$\mathbf{EMA}_{22}$	σ_{5d}	σ_{22d}	$\ln(\mathbf{VXN})$	$\ln(\sigma_{\mathbf{Parkinson}})$
9	✓	✓	✓	✓	✓	✓	✓	✓	✓
8	✓	✓	✓	✓	✓	✓	✓	✓	
7	✓	✓	✓	✓	✓		✓	✓	
6	✓	✓	✓	✓	✓		✓		
5	✓	✓	✓	✓	✓				
4		✓	✓	✓	✓				
3		✓	✓	✓					
2		✓		✓					
Final	✓	✓	✓	✓	✓		✓		

Table 19 presents the initial feature set considered for the GMM analysis and the final feature set selected for regime identification. Interestingly, the inclusion of ln(VXN) does not improve the model's predictive performance. This suggests that VXN provides limited incremental information for distinguishing between volatility regimes once historical volatility measures are incorporated, particularly the 1-day realized volatility measure. The result may indicate that recent observed volatility already captures much of the information relevant for regime identification, reducing the additional contribution of the forward-looking volatility expectations reflected in VXN. Therefore, the final regime classification is driven primarily by recent and historical volatility dynamics rather than by the additional information contained in VXN. The ability of the final feature set to produce distinct regimes is presented in Appendix D, where the validation-set results demonstrate that the model performs well in separating observations into distinct regimes.

Table 20: Sequential evaluation of GMM feature dimension reduction

Model	**Features**	**Mean Log-Likelihood**	**Model**	**Features**	**Mean Log-Likelihood**
Model 9	9	−0.6706	Model 5	5	1.5153
Model 8	8	0.2288	Model 4	4	0.4993
Model 7	7	0.8741	Model 3	3	0.3366
Model 6	**6**	**1.6572**	Model 2	2	−0.5224

Based on the mean log-likelihood criterion, the six-feature specification achieves the best performance and is therefore selected as the final GMM model for estimating regime probabilities and identifying volatility regimes.

The regime-specific feature means reported in Table 21 show a consistent pattern across the training, validation, and test sets. Regime 0 is characterized by negative values across all volatility-related features, indicating relatively calm market conditions, whereas Regime 1 exhibits positive values across all features, corresponding to periods of elevated volatility and

Table 21: Comparison of standardized feature means across GMM regimes for Training, Validation, and Test sets

Regime	RV_{1d}	RV_{5d}	RV_{22d}	EMA_5	EMA_{22}	σ_{22d}
Training Set						
Regime 0	-0.5638	-0.5669	-0.5623	-0.5280	-0.5347	-0.5097
Regime 1	1.1927	1.1992	1.1896	1.1169	1.1311	1.0783
Validation Set						
Regime 0	-0.3369	-0.3387	-0.2867	-0.4025	-0.3753	-0.4349
Regime 1	0.8206	0.8279	0.8328	0.5466	0.5583	0.5839
Test Set						
Regime 0	-0.3973	-0.4022	-0.4001	-0.4507	-0.4521	-0.4167
Regime 1	0.8261	0.8424	0.7744	0.6344	0.5563	0.4714

crisis-like market conditions. This consistency across the three datasets suggests that the GMM provides a stable and economically interpretable regime classification.

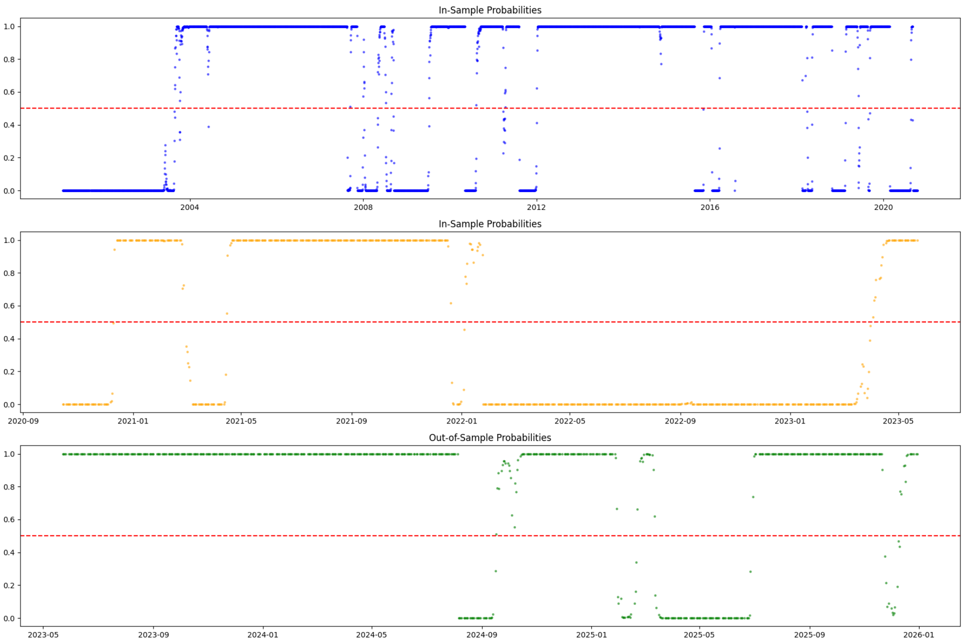

Figure 10: Predicted regime probabilities across train, validation, and test samples.

Figure 10 illustrates the predicted posterior probabilities of belonging to Regime 0. The probabilities are equal to 1 or 0 for most observations, indicating that the model assigns Regime 0 with very high confidence during these periods. Only a small number of observations lie close to the dotted red line representing a probability of 0.5, suggesting limited uncertainty in regime classification. This is further quantified by the ambiguity rate, defined as the percentage of trading days for which the posterior probability of belonging to Regime 1, $P(S_t = 1)$, falls strictly between 0.40 and 0.60, as presented below.

Table 22: GMM regime assignment ambiguity rates across train, validation, and test samples

Sample Subperiod	Evaluation Context	Ambiguity Rate ($0.4 \le P(S_t = 1) \le 0.6$)
Training Set	In-Sample	0.60%
Validation Set	In-Sample	0.76%
Test Set	Out-of-Sample	0.61%

As shown in Table 22, the ambiguity rates are consistently low across the training, validation, and test sets, remaining below 1.0% in all cases. This indicates clear separation between the two GMM components and suggests that regime assignments are associated with relatively low classification uncertainty.

6.5 Regime-Aware Ensemble Results

The regime-aware ensemble combines the standalone LSTM with the best-performing AR-GARCH-Skew-t specification, which uses an AR(1) mean equation, a standard GARCH(1,1) volatility model, and skewed Student's t innovations. The relative weights of the two forecasts are determined by the GMM-estimated regime probabilities, allowing the ensemble to adapt to different volatility conditions.

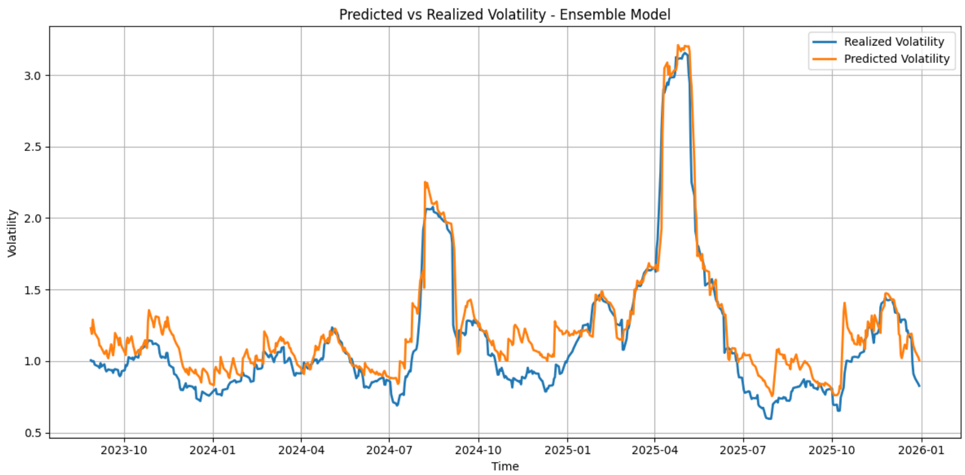

Figure 11: Predicted and Realized Volatility - Ensemble Model.

Figure 11 shows that the regime-aware ensemble performs worse than the standalone LSTM, particularly during low-volatility periods. This underperformance appears to be driven primarily by the relatively weak forecasts of the GARCH component rather than by the regime probabilities generated by the GMM. When compared with major market-stress episodes, the GMM appears to identify high-volatility periods reasonably well. For example, the volatility increase in August 2024, associated with the global market sell-off, concerns about U.S. economic growth, and the unwinding of the yen carry trade, is correctly captured. Similarly, the elevated volatility observed from February to May 2025, amid growing concerns surrounding

AI valuations and technology stocks, is also identified as a high-volatility period. These results suggest that the regime identification mechanism itself performs reasonably well; however, incorporating the GARCH forecasts into the ensemble does not provide additional forecasting value. In particular, the relatively weak GARCH performance during low-volatility periods limits the overall effectiveness of the regime-aware ensemble.

6.6 Overall Forecast Comparison

In the final stage, the out-of-sample forecasting performance of all considered models is evaluated jointly to assess their relative predictive accuracy. The comparison is based on QLIKE, MAE, and RMSE across the baseline models, GARCH-family models, standalone LSTM, hybrid LSTM, and regime-aware ensemble models.

Table 23: Out-of-sample forecast accuracy metrics across all evaluated models

Model	**RMSE**	**MAE**	**QLIKE**
Simple Exponential Smoothing	0.0602	0.0293	1.0739
Hybrid LSTM	0.0842	0.0514	1.0749
Standalone LSTM	0.0909	0.0519	1.0751
Ensemble Model	0.1598	0.1227	1.0835
GARCH(1,1)	0.2015	0.1599	1.0860
Simple Moving Average	0.3026	0.1938	1.0922
Historical Average	0.5345	0.4437	1.1520

Table 23 summarizes the out-of-sample performance of all considered models, with the results ranked according to QLIKE. Simple Exponential Smoothing achieves the lowest QLIKE and therefore the best overall performance under this criterion. The standalone LSTM and regime-aware ensemble also show relatively similar QLIKE values. Compared with the Simple Moving Average and Historical Average benchmarks, the developed models generally achieve better forecasting performance. To determine whether the observed differences are statistically significant, pairwise Diebold–Mariano tests are conducted, with the results presented in Figure 12.

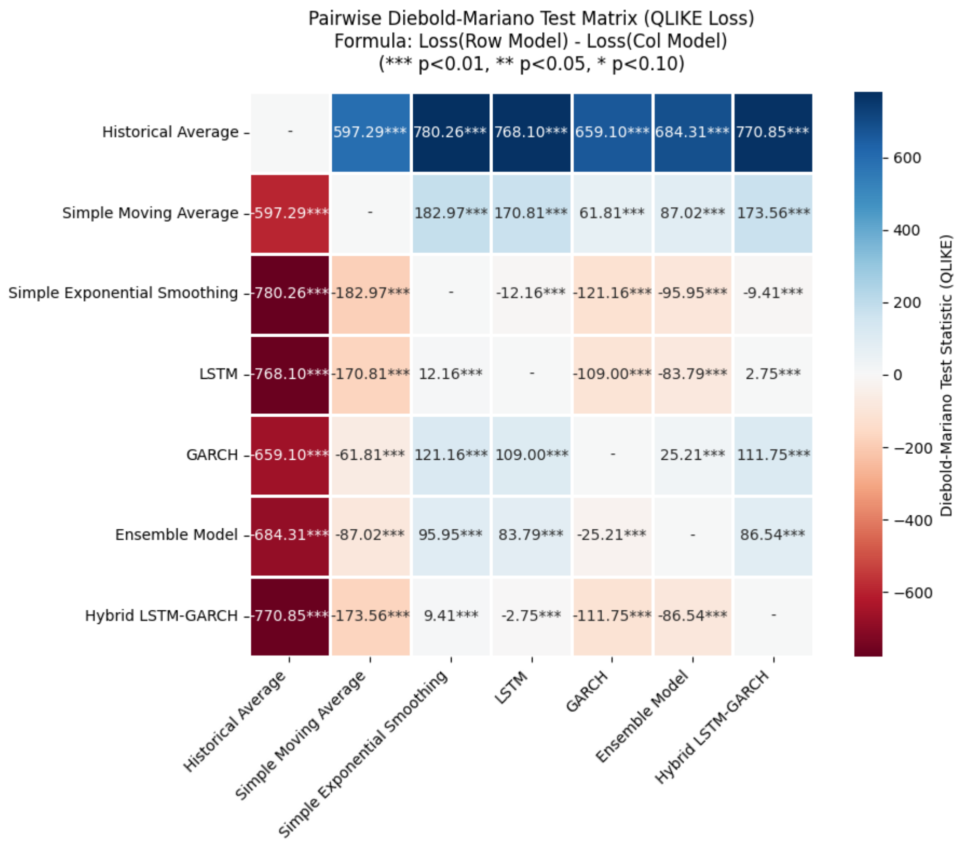

Figure 12: Pairwise Diebold–Mariano Tests Based on QLIKE Loss.

The pairwise Diebold–Mariano test results indicate that the differences in forecasting performance between the models are statistically significant at the 1% significance level. In Figure 12, a negative value in a cell indicates that the model in the corresponding row outperforms the model in the corresponding column, as it achieves a lower QLIKE loss. Conversely, a positive value indicates that the row model has a higher loss and therefore performs worse than the column model.

The results show that the Simple Exponential Smoothing model consistently outperforms the other models under the QLIKE loss criterion, dominating the pairwise comparison matrix. The hybrid LSTM also performs better than the standalone LSTM, indicating that incorporating GARCH-based volatility forecasts as additional inputs can improve the forecasting performance of the LSTM.

The GARCH-family models outperform only the two simplest benchmarks, the Historical Average and Simple Moving Average, while remaining inferior to Simple Exponential Smoothing and the LSTM-based models. This result may be related to the characteristics of the Nasdaq-100 realized volatility series discussed in the preceding sections. The series exhibits substantial short-term fluctuations and remains highly jagged even during relatively stable market periods. Such rapid changes in volatility dynamics may limit the ability of the GARCH models to adjust sufficiently quickly to new market conditions, particularly when volatility changes occur over short horizons. This limitation is reflected in their relatively weaker out-of-sample performance and, consequently, reduces the potential benefit of incorporating GARCH forecasts into the regime-aware ensemble.

The strong persistence in volatility identified by the GARCH analysis may help explain why Simple Exponential Smoothing achieves the best overall forecasting performance. SES places greater weight on recent observations while retaining information from the historical volatility series, allowing it to respond relatively quickly to changes in the current volatility level without explicitly modelling complex volatility dynamics. This characteristic may be particularly advantageous for the Nasdaq-100 realized volatility series, which exhibits substantial short-term fluctuations and frequent changes in volatility. In contrast, LSTM models attempt to learn more complex nonlinear temporal relationships from historical observations. Their greater model complexity and larger number of estimated parameters may also make their forecasts more sensitive to the choice of inputs, initialization, hyperparameters, and training sample.

The superior QLIKE performance of SES may also be related to its response to sharp volatility increases. QLIKE is asymmetric and assigns a particularly large loss to forecasts that substantially underestimate realized volatility. Models that produce smoother forecasts may therefore incur relatively large QLIKE losses during sudden volatility spikes. In contrast, SES, by assigning greater weight to recent observations, can adjust its forecasts more rapidly following an increase in realized volatility. This may allow it to avoid some of the large losses associated with underprediction and helps explain its strong performance under the QLIKE criterion.

7 SUMMARY

The study evaluated whether integrating econometric and deep learning approaches could improve the forecasting of Nasdaq-100 realized volatility. Two integration strategies were examined: feature-level integration through a hybrid LSTM and prediction-level integration through a regime-aware ensemble using GMM-derived regime probabilities. Their out-of-sample performance was compared with GARCH-family models, a standalone LSTM, and three conventional benchmarks using QLIKE, MAE, and RMSE, with Diebold–Mariano tests used for statistical comparison.

The first hypothesis concerned the performance of the developed forecasting models relative to the benchmark models. The results provide partial support for H1. The developed models generally outperform the Historical Average and Simple Moving Average benchmarks; however, SES achieves the lowest QLIKE and therefore the best overall performance. This finding suggests that greater model complexity does not necessarily improve volatility forecasting, while SES may benefit from its ability to adapt quickly to short-term changes in the Nasdaq-100 volatility level.

The results support H2, with the standalone LSTM outperforming the individual GARCH-family models in forecasting performance. This suggests that the nonlinear sequence-learning capabilities of the LSTM can provide additional predictive value when modelling the highly variable Nasdaq-100 volatility series. The GARCH-family models, although able to capture important characteristics of volatility such as persistence and conditional heteroskedasticity, appear less responsive to rapid changes in realized volatility. Their forecasts are generally smoother and tend to react with a delay to sharp volatility movements. This limitation is reflected in their relatively weaker out-of-sample performance. Nevertheless, the GARCH models still outperform the Historical Average and Simple Moving Average, demonstrating that the traditional parametric models retain useful forecasting information despite their limitations.

The findings support H3, which examined whether incorporating GARCH forecasts into the LSTM improves its forecasting performance. The hybrid LSTM performs better than the standalone LSTM, indicating that the forecasts generated by the GARCH-family models contain information that is useful to the neural network. This improvement suggests that GARCH forecasts may capture aspects of the conditional volatility dynamics that complement the information learned directly from the historical data by the LSTM. The improvement of the hybrid model therefore provides evidence that combining parametric and deep learning approaches can be beneficial when the information from the two model classes is incorporated appropriately.

The GMM provides a useful probabilistic representation of changing volatility conditions, supporting its application for identifying market regimes within the proposed forecasting framework. Despite the successful regime identification, the results do not support H4. The regime-aware ensemble, which combines the best-performing GARCH model and the standalone LSTM using GMM-derived regime probabilities, does not outperform the hybrid LSTM.

These results suggest that the performance of the ensemble model depends strongly on the forecasting quality of its individual components, in addition to the method used to combine them. The relatively weak forecasting performance of the GARCH component is likely an important factor in this result. Although the GMM can identify changes in volatility conditions, assigning greater weight to a weaker forecasting component under a particular regime does not necessarily improve the overall forecast.

The comparison with the existing literature provides both supporting and contrasting evidence. The improvement of the hybrid LSTM over the standalone LSTM is broadly consistent with previous research showing that GARCH-based information can provide complementary information to neural networks in volatility forecasting. The results are also consistent with studies emphasizing that model complexity does not automatically result in superior forecasting performance and that simple benchmarks should remain part of rigorous forecasting evaluations. However, the present study extends these findings by comparing the hybrid approach with simple forecasting benchmarks and by examining a separate regime-aware ensemble. In particular, the finding that Simple Exponential Smoothing achieves the best overall QLIKE performance demonstrates that the additional complexity of GARCH, LSTM, and hybrid architectures does not necessarily translate into lower forecasting losses for the Nasdaq-100 realized volatility series. This result is consistent with Tse and Tung (1992), who found that EWMA outperformed GARCH in forecasting volatility in the Singapore stock market, as well as with Makridakis, Spiliotis, and Assimakopoulos (2018), who reported that simple statistical methods can outperform more complex machine-learning approaches in out-of-sample forecasting. Similarly, Romero and Opschoor (2026) find that the EWMA model cannot be consistently outperformed by more complex high-frequency-based volatility models at daily and weekly forecasting horizons, while remaining competitive at the monthly horizon. These results further highlight the continued forecasting value of simple, parsimonious volatility models.

The regime-aware ensemble results provide a mixed picture relative to the existing literature as discussed in the literature review. While previous studies suggest that incorporating regime information can improve volatility forecasts in some settings, other research reports limited out-of-sample gains from regime-switching approaches. The present study extends this literature by using GMM-derived regime probabilities to combine GARCH and LSTM forecasts. However, despite clear separation between calm and crisis regimes, the regime-aware ensemble does not outperform the hybrid LSTM. This suggests that accurate regime identification alone is insufficient to improve forecasting performance. Instead, the usefulness of a forecasting approach depends on both the information captured by its components and the way this information is integrated.

Overall, the Nasdaq–100 realized volatility series exhibits strong persistence, heavy tails, substantial short-term fluctuations, and pronounced volatility spikes, creating a challenging forecasting environment. The GARCH–family models capture persistent volatility dynamics but produce relatively smooth forecasts, while the LSTM offers greater flexibility in capturing nonlinear temporal patterns. The hybrid architecture benefits from incorporating GARCH-based information at the feature level, allowing the LSTM to learn how this information interacts with historical volatility patterns, whereas the regime-aware ensemble does not provide additional gains when combining forecasts at the prediction level. This suggests that integrating complementary information within the forecasting model may be more effective than combining model outputs after predictions have already been generated. Ultimately, SES achieves the best overall forecasting performance, demonstrating that a simple and adaptive approach can outperform more complex models when recent observations contain the most useful predictive information.

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Appendix A: Autocorrelation Diagnostics for Model Residuals

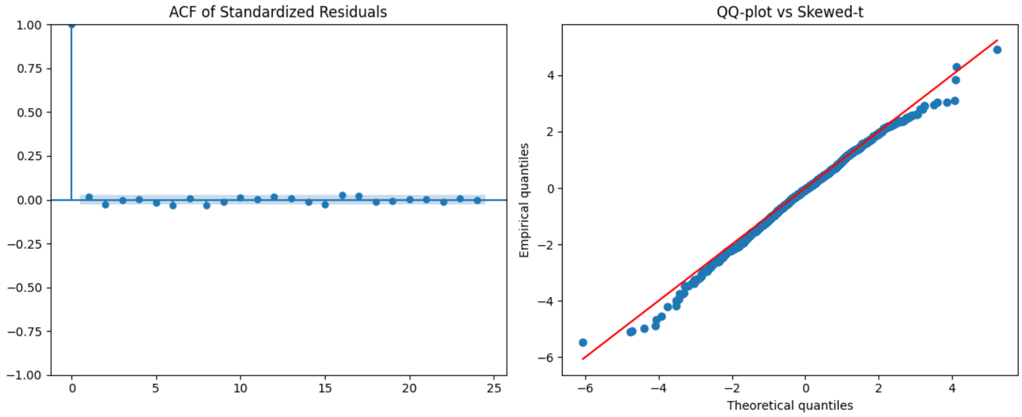

Figure A.1: ACF Plots of Standardized Residuals for the ARMA(1,0)–GARCH(1,1) Model with a Skewed Student's *t* Distribution

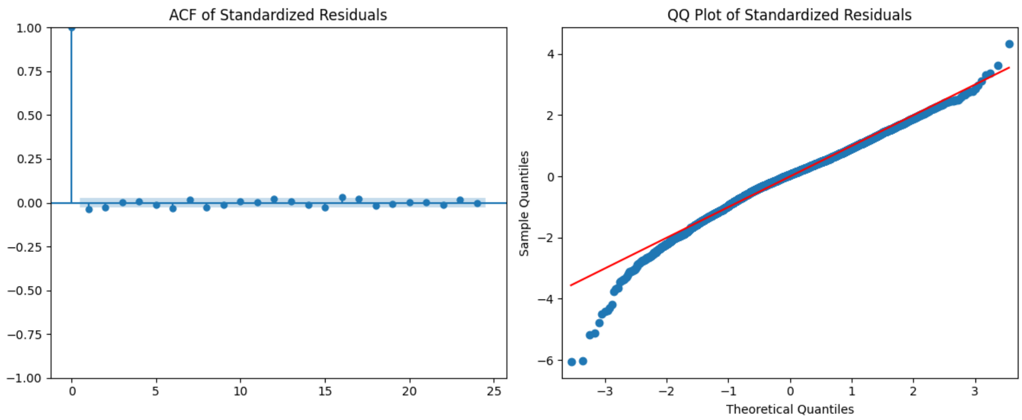

Figure A.2: ACF Plots of Standardized Residuals for the EGARCH(1,1) Model

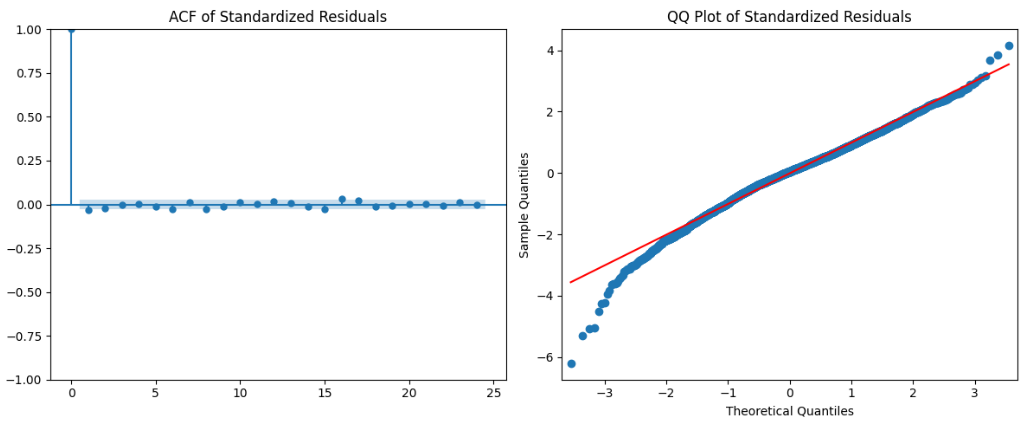

Figure A.3: ACF Plots of Standardized Residuals for the GJR-GARCH(1,1) Model

Appendix B: Parameter Stability and Rolling-Window Diagnostics

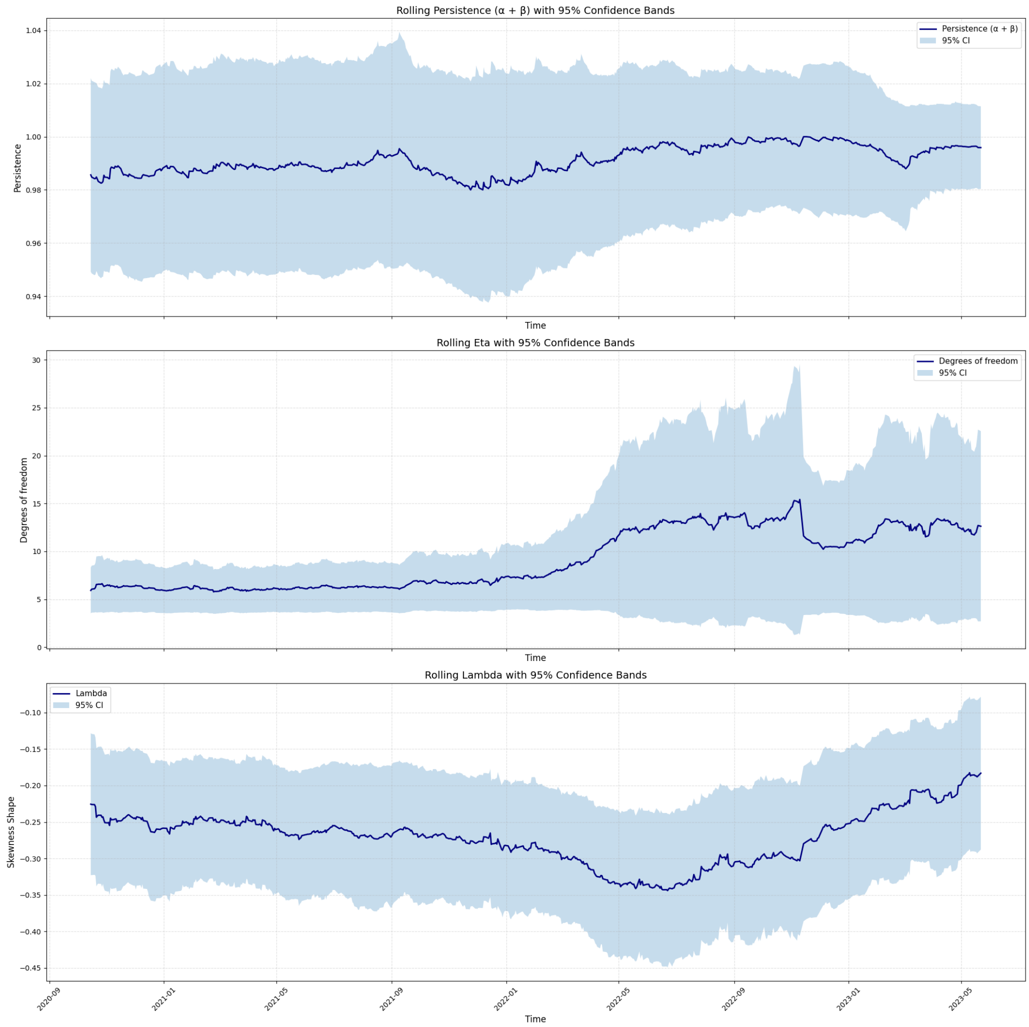

Figure B.1: Parameter Stability and Rolling-Window Diagnostics for the ARMA(1,0)–GARCH(1,1) Model with a Skewed Student's t Distribution

Appendix C: Correlation Matrices of Initial Input Features

C.1 Standalone LSTM Model

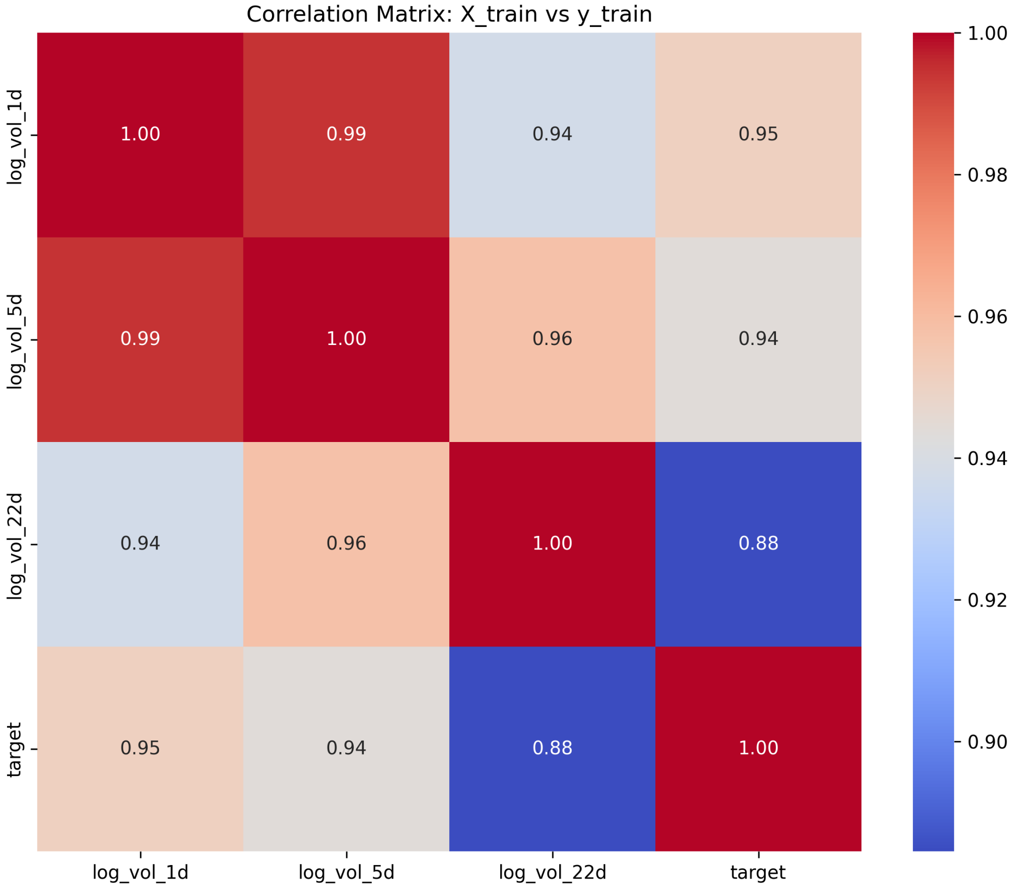

Figure C.1: Correlation Matrix of Initial Input Features for Standalone LSTM

C.2 Hybrid Model

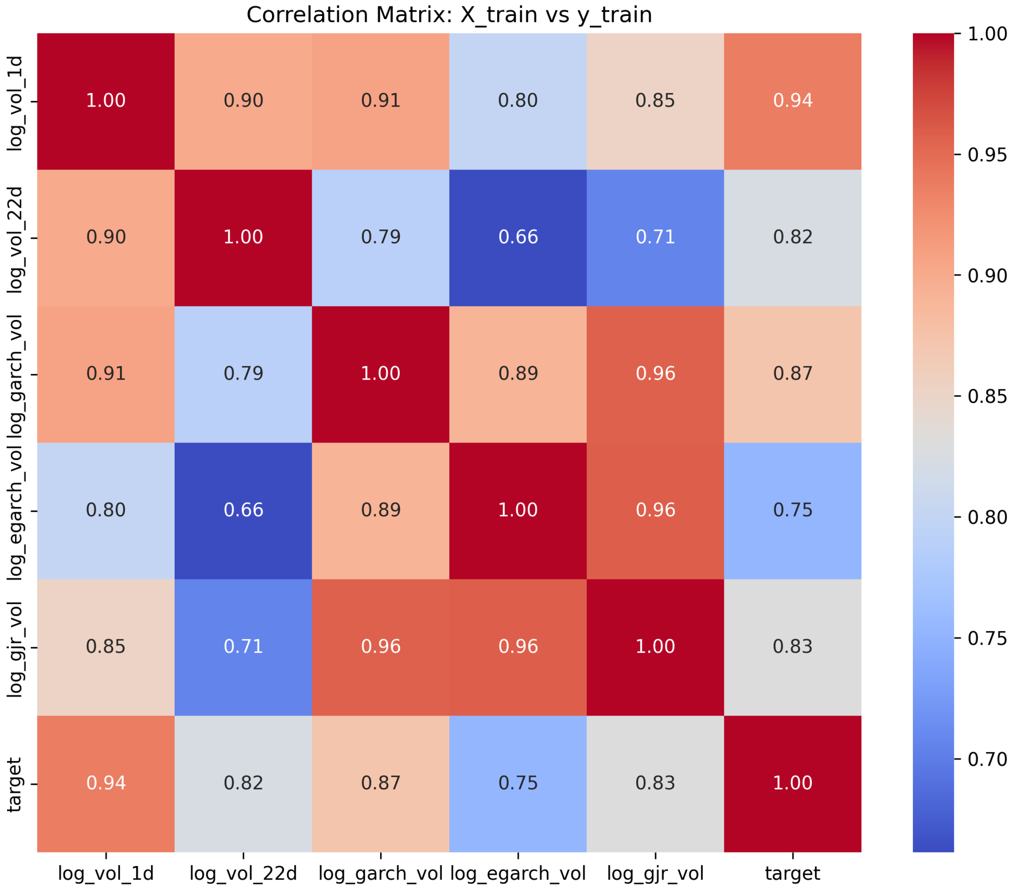

Figure C.2: Correlation Matrix of Initial Input Features for Hybrid Model

Appendix D: Box Plots of Input Features by GMM-Derived Regimes

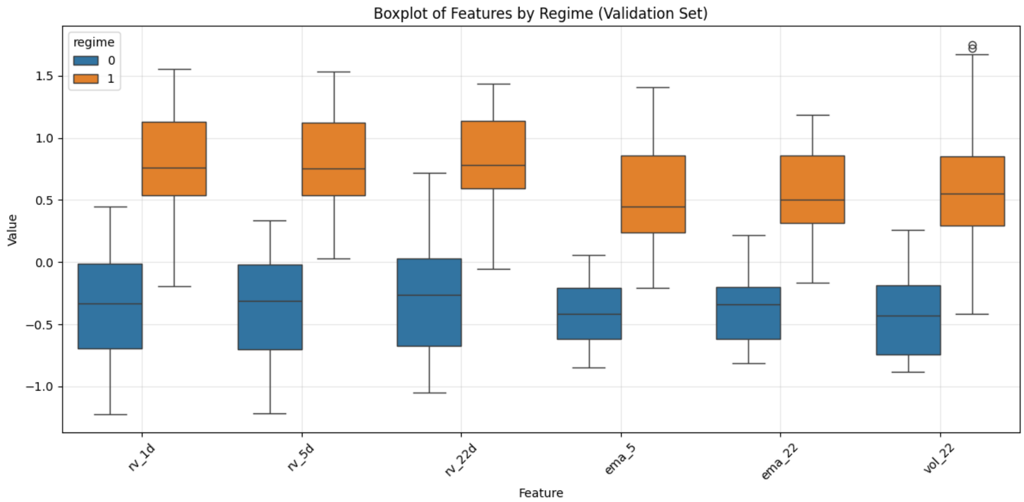

Figure D.1: Distribution of Input Features Across GMM-Derived Regimes (Validation Sets)